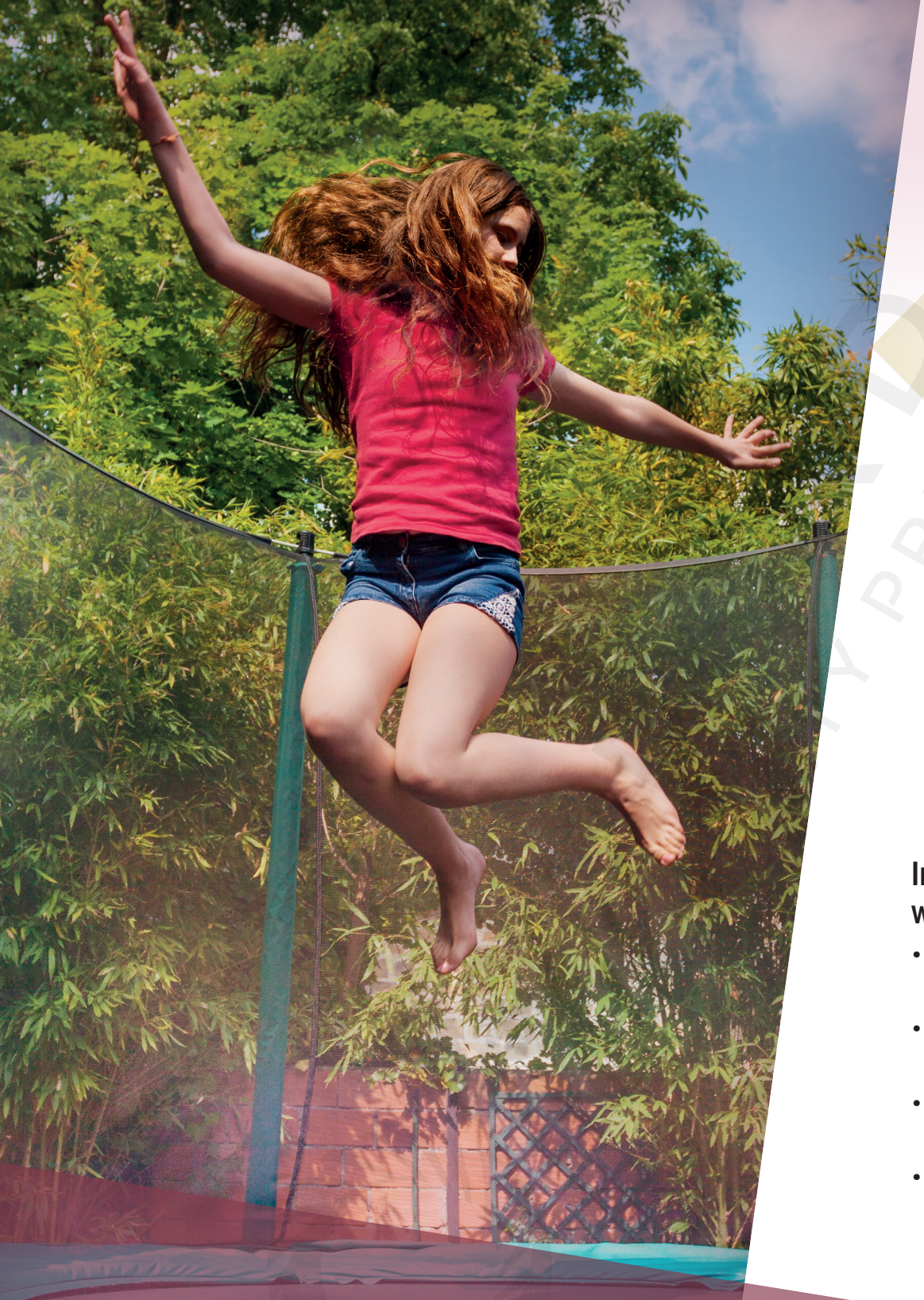


6

In this chapter, we will learn about

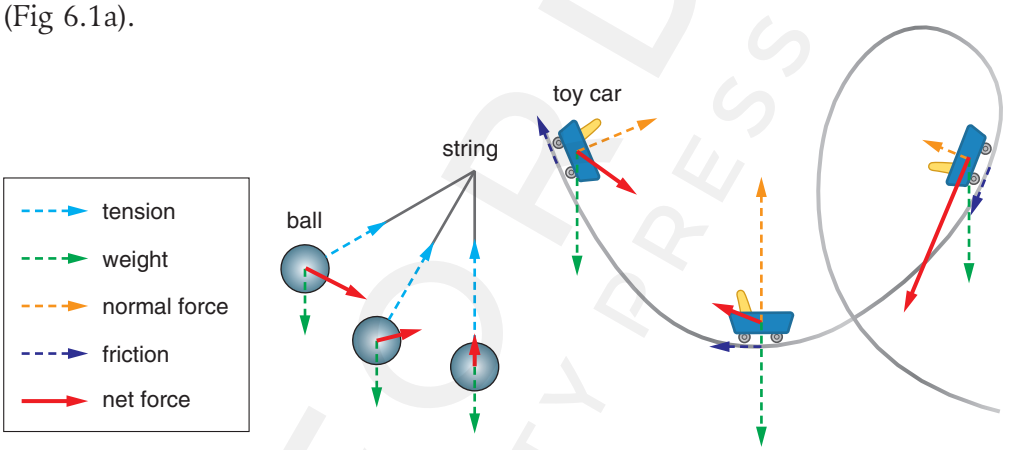
- work as a way of energy transfer
- kinetic energy and potential energy
- the law of conservation of energy in a closed system
- power as the rate of energy transfer

Work, Energy and Power

6.1 Work

Flipped classroom
Watch a video and answer the questions.

Newton's second law of motion ($F = ma$) can be used to analyse the motion of objects as long as the objects have constant acceleration, i.e. they are acted on by a constant net force. However, in real situations, the net force is usually not constant throughout the motion of an object (Fig 6.1a).



(i) A swinging ball. *(ii) A toy car moving on a track.*

Fig 6.1a *Motion where the net force is not constant.*

In this chapter, we will learn to tackle the problems involving varying net force using the energy approach. To start with, we need to learn about the concept of work.

1 Work: another way of energy transfer

DSE goal

Interpret mechanical work as a way of energy transfer.

Define mechanical work done $W = Fs\cos\theta$.

Solve problems involving mechanical work.

DSE exam

22(1A)Q10

Stress that energy is converted from one form to another whenever work is done.

From Book 1 Chapter 2, we know that heat is the energy transferred due to a temperature difference. The concept of work is similar to heat.

Work is the energy transferred due to the action of a force over a displacement.

Consider a force F acting on an object over a displacement s. The **work** (or **mechanical work**) W done by F is defined as

Work = force parallel to displacement × displacement

$$W = F_{/\!/}\, s$$

Similar to heat, work is not an intrinsic property of an object. We cannot say that an object gains a certain amount of work.

work 功　　mechanical work 機械功

In the special case that the force acts in the same direction as the displacement (Fig 6.1b),

In this case, $F_{//} = F$. ▶

$$W = Fs \quad \text{..... (1)}$$

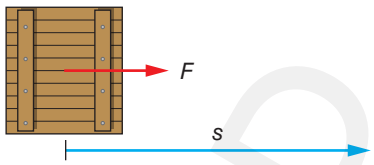
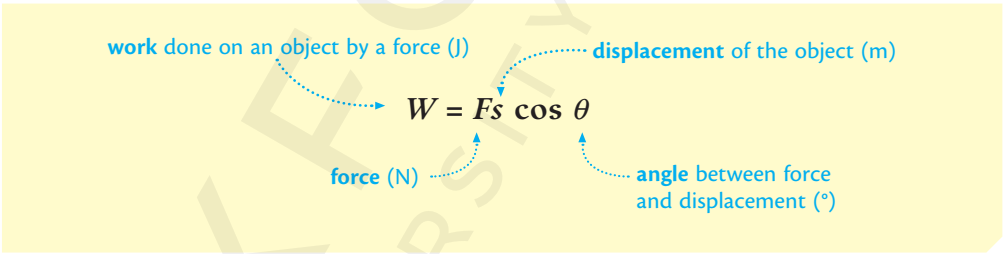


Fig 6.1b *Force acts in the same direction as the displacement.*

However, in general cases, the force and the displacement make an angle θ. Then, only the component of force, $F \cos \theta$, which is parallel to the displacement, will do work (Fig 6.1c). As a result,

$$W = (F \cos \theta) \times s$$

i.e.



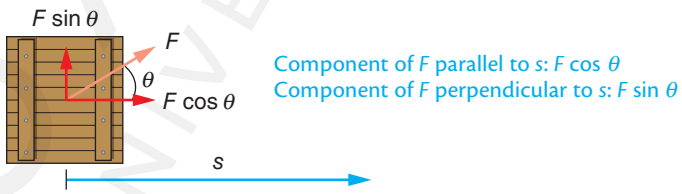


Fig 6.1c *Force and displacement making an angle θ.*

The unit of moment is also N m, in which the metre term comes from the moment arm. It is not recommended to express work in terms of N m to avoid confusion as work and moment are two different physical quantities.

The SI unit of work is the joule, written as J. One joule of work is done when a force of 1 N moves a displacement of 1 m in the direction of the force.

$$\mathbf{1\ J = 1\ N\ m}$$

Note that:

1. Work is a scalar quantity: it has magnitude but no direction.
2. When the force acts in the same direction as the displacement, $\theta = 0°$ and $\cos 0° = 1$. The equation becomes $W = Fs$.
   When the force is perpendicular to the displacement, $\theta = 90°$ and $\cos 90° = 0$. The equation becomes $W = 0$.

This is equation (1) ▶

Therefore, work is not done in the following cases (Table 6.1a).

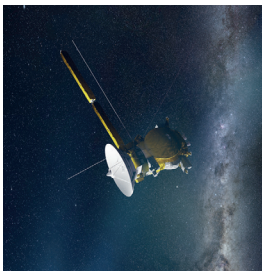
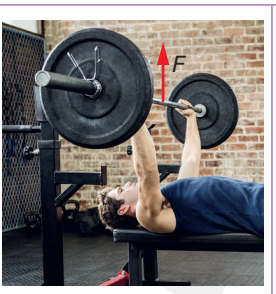
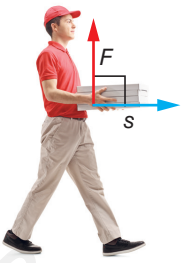
Cases			
	No work is done on a spacecraft moving in outer space at a constant velocity.	No work is done on the barbell held still by a weightlifter.	No work is done on the boxes held by a man who walks at a constant velocity.
Reasons	$F = 0$	$s = 0$	$\theta = 90°$ ($\cos\theta = 0$)

Table 6.1a *Examples where work is not done.*

Know more

Holding a heavy object still

Why do you feel tired when holding a heavy object still for some time?

▶ p.276

Another example to illustrate this: a car crashes into a wall and stops. Although the wall exerts a large force on the car, it does no work since its point of application does not move. The car consists of a large number of particles. When the car deforms in the collision, the forces between the particles do work and the KE of the car changes into internal energy.

Example 1 Work done on a toy

A girl pulls a toy for 0.1 m with a horizontal force of 2 N (Fig a). A boy then pulls the toy for another 0.1 m with a force of 2 N, but his force makes an angle of 30° to the horizontal (Fig b).



Fig a

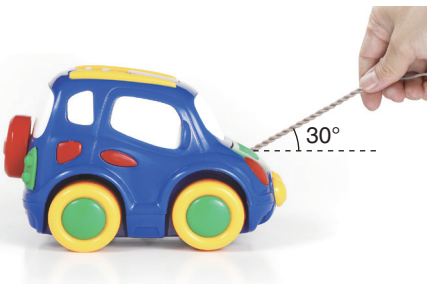


Fig b

(a) Find the work done by the girl and the boy on the toy.

(b) The boy releases the toy and it moves 5 cm forwards before it stops. What is the work done by the boy on the toy in this part of journey?

Solution

(a) Apply $W = Fs\cos\theta$.

Work done by the girl $= 2 \times 0.1 \times \cos 0°$

$= 0.2$ J

Work done by the boy $= 2 \times 0.1 \times \cos 30°$

$= 0.173$ J

(b) 0

Tip

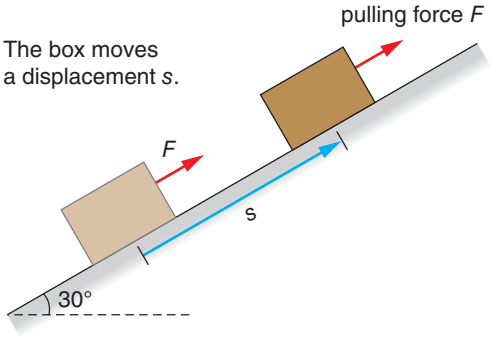
The force acting on the toy by the boy is zero in this part of journey. ▶

Google Forms Microsoft Forms

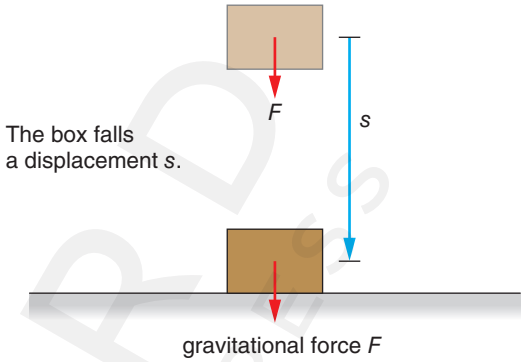
Quick check 1

1 Determine the work done on the box by the force F in each of the following cases.

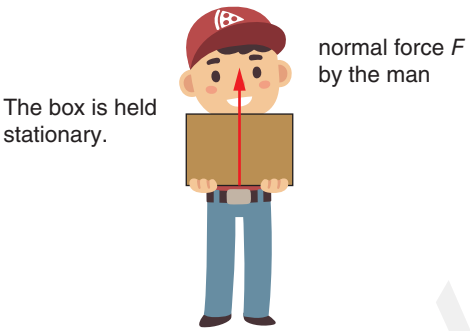
(a)



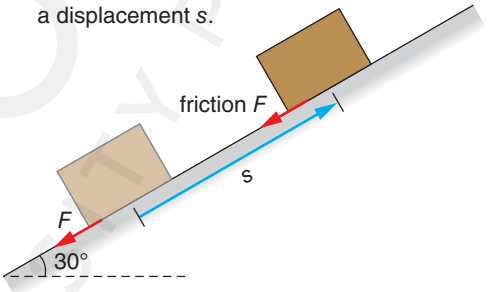
(b)



(c)



(d)



Learn by practice 1

1 Ivy pushes a cart with a force of 50 N over a displacement of 10 m in the same direction as the force. How much work does she do on the cart?

2 Ken holds a box with an upward force of 100 N (Fig a) and walks 8 m to the right. What is the work done by the upward force?



Fig a

3 John pulls a 10-kg sledge for 1.6 m with a force of 15 N at 50° to the horizontal (Fig b). How much work does he do on the sledge?

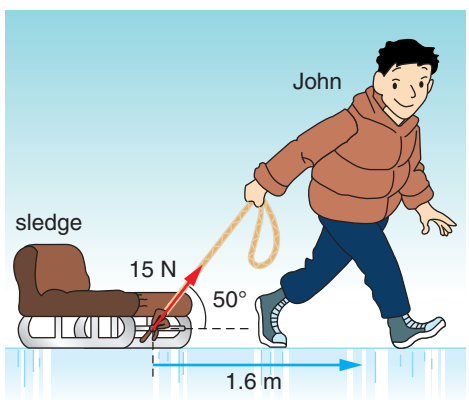


Fig b

2 Positive and negative work

Work can be positive or negative. The sign of work indicates whether an object gains or loses energy.

When an applied force acts in the same direction as the displacement, the force does **positive** work on the object and the object gains energy (Fig 6.1d).

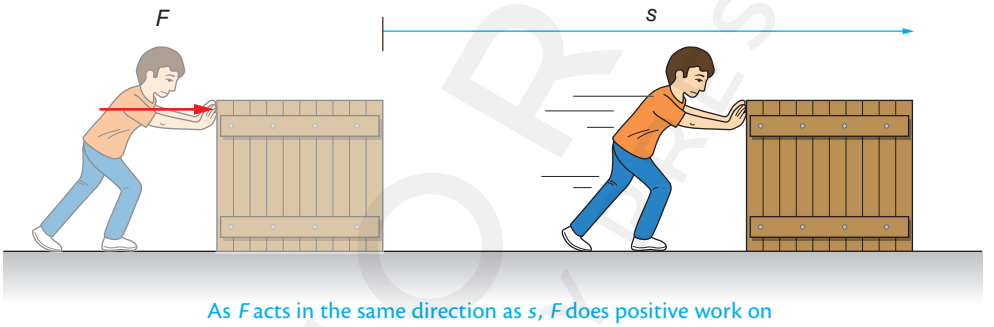


Fig 6.1d *Doing positive work.*

When the applied force acts in the opposite direction to the displacement, the force does **negative** work on the object and the object loses energy. In Figure 6.1e, a moving object is acted on by a friction which is opposite to the object's displacement. We say that the work done by the friction on the object is –18 J. Alternatively, we may also say that the **work done against** friction is 18 J so that we can leave out the negative sign.

In other words, work done against a force always causes an object to lose energy. ▶

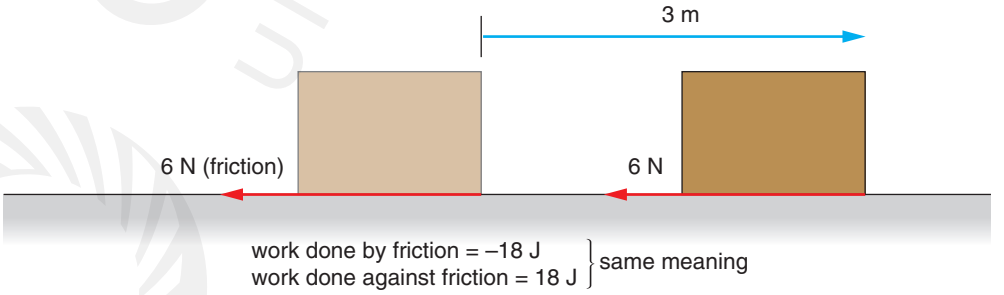


Fig 6.1e *Work done against a force.*

If the applied force makes an angle with the displacement, only the component parallel to the displacement will do positive or negative work on the object.

3 Work done by more than one force

Very often, an object is acted on by several forces. The total work done on the object can be found by summing up the work done by all the forces. It can also be obtained by calculating the work done by the net force. Figure 6.1f shows an example.

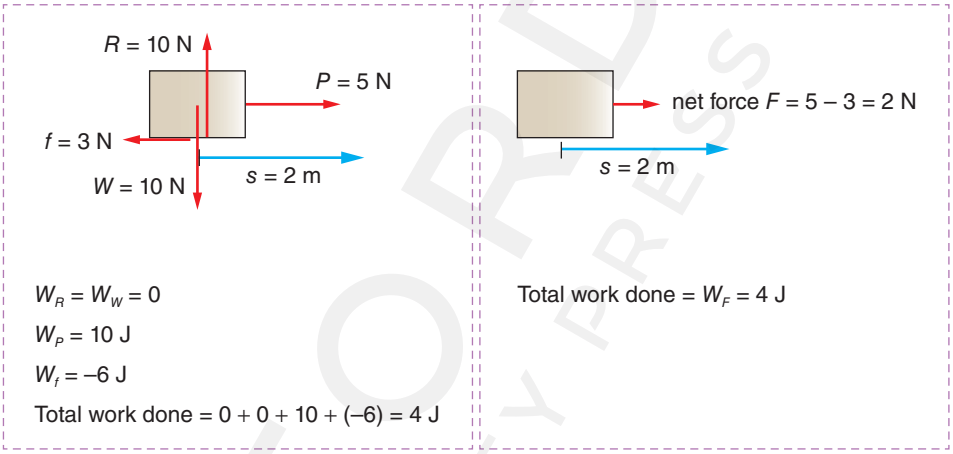
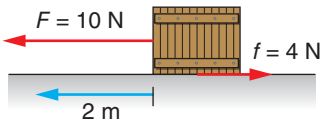


Fig 6.1f *Finding total work done.*

Example 2 Change in energy of an object

A force F of 10 N acts on an object over a displacement of 2 m towards the left on a rough horizontal surface as shown. The friction f acting on the object is 4 N. What is the change in energy of the object?



Solution

Net force along displacement = 10 + (–4) = 6 N (towards the left)

Work done by net force = $Fs = 6 \times 2 = 12$ J

$\therefore$ The object gains 12 J of energy.

Learn by practice 2

1 A 3-kg box is pushed by a horizontal force of 10 N to move 5 m to the right on a rough horizontal surface (Fig a). The friction acting on the box is 2 N.

(a) Find the work done **by** the pushing force.

(b) Find the work done **against** friction.

(c) Find the change in energy of the box.

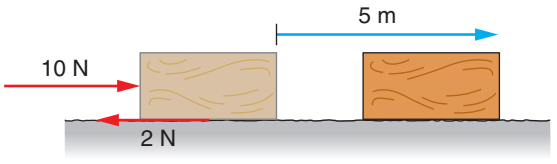


Fig a

Exercise 6.1

1 Which of the following cannot be a unit of work?

A J

B $J\ s^{-1}$

C N m

D $kg\ m^2\ s^{-2}$

2 Which of the following forces does **no** work on the box?

A The force pulling a box to move it on a rough floor

B The tension in a cable pulling a box to move upwards

C The tension in a string hanging a box which remains at rest

D The friction acting on a box sliding down a slope

***3** Which of the following statements about work is/are correct?

(1) It is a scalar.

(2) It is the energy stored in an object when it moves.

(3) It is the energy transfer due to a force.

A (2) only

B (1) and (2) only

C (1) and (3) only

D (1), (2) and (3)

4 A net force F of 5 N acts on an object so that the object moves a displacement s of 2 m. F makes an angle θ with s (Fig a).

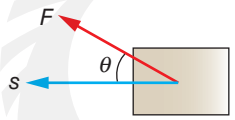


Fig a

Find the work done by F and state whether the object gains or loses energy in each of the following cases.

(a) $\theta = 60°$

(b) $\theta = 100°$

***5** A force F of 9 N acts on a block over 3 m on a rough horizontal surface. The friction f acting on the block is 5 N (Fig b).

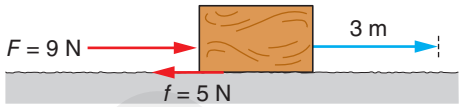


Fig b

(a) Find the work done **by** F and f respectively.

(b) What is the meaning of the sign of work?

(c) What is the change in energy of the block?

***6** A force of 8 N pulls a block to the right on a rough horizontal surface (Fig c). The friction acting on the block is 4 N.

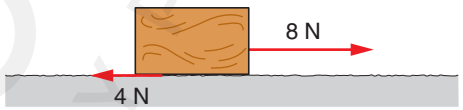


Fig c

After the block moves a distance s, the total work done on the block is 12 J.

(a) Find s.

(b) The weight of the block and the normal force acting on the block by the surface do no work on the block. Why?

***7** A box is placed on a rough inclined plane (Fig d). After it is given a sharp push, it moves up the plane. The weight W of the box is 5 N. The friction f acting on the box is 1 N. The normal force R acting on the box by the plane is 4.33 N.

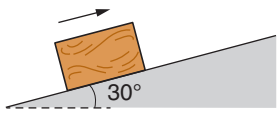


Fig d

The box moves 0.5 m up the plane.

(a) Find the work done **by** each of W, f and R.

(b) What is the change in energy of the block?

6.2 Mechanical energy

Kinetic energy and potential energy are collectively called **mechanical energy**. We will study these forms of energy in detail.

DSE goal

State that kinetic energy is the energy possessed by an object due to its motion.

Derive KE = $\frac{1}{2}mv^2$.

Solve problems involving kinetic energy.

1 Kinetic energy

Kinetic energy (KE) is **the energy possessed by an object due to its motion** (Fig 6.2a). The object possesses more KE when it moves faster. You will learn to derive a relationship relating KE and speed below.



Fig 6.2a *An object does not possess KE when it is stationary. It possesses KE when it moves.*

Simulation

→ This simulation allows Ss to apply a force to push a toy car on a frictionless track and find out how the work done on the car is related to the change in kinetic energy of the car.

In Figure 6.2b, a constant net force F acts on a stationary object. This makes the object move at a uniform acceleration a for a displacement s parallel to F. The velocity of the object increases from 0 to v.

Since the acceleration a is uniform,

$$v^2 = \cancel{u^2}^{0} + 2as \implies s = \frac{v^2}{2a}$$

By $W = Fs$ and $F = ma$, the work done by F can be written as:

$$W = Fs = (m\cancel{a})\left(\frac{v^2}{2\cancel{a}}\right) = \frac{1}{2}mv^2$$

where m is the mass of the object.

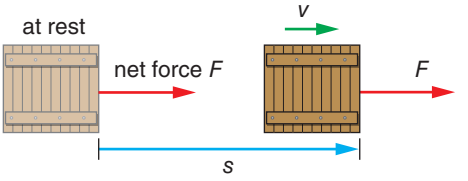


Fig 6.2b *An object acted on by a force F accelerates for a displacement s.*

The work done by F appears as the KE of the object. Therefore, we define the KE of an object as

$$\text{KE} = \frac{1}{2}mv^2$$

kinetic energy of object (J); mass of object (kg); speed of object (m s^{-1})

mechanical energy 機械能

Note that:

1 The KE of an object depends on both its mass m and its speed v. It is directly proportional to m and v^2 (Fig 6.2c and 6.2d). It doubles when m doubles, and increases to 4 times when v doubles (Fig 6.2e).

2 KE cannot be negative. It is a scalar and it does not depend on the direction of motion.

3 If the speed of an object changes from u to v, the change in KE (ΔKE) is given by

$$\Delta\text{KE} = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}m(v^2 - u^2)$$

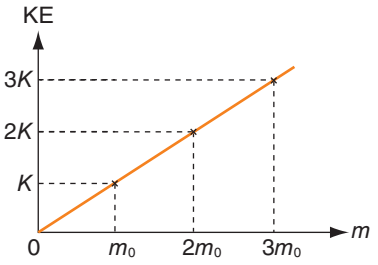


Fig 6.2c *Graph of KE against m for constant v.*

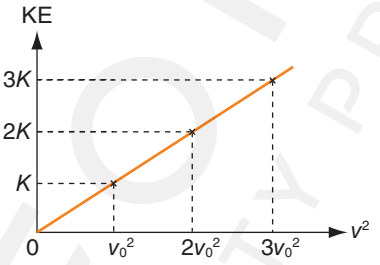


Fig 6.2d *Graph of KE against v^2 for constant m.*

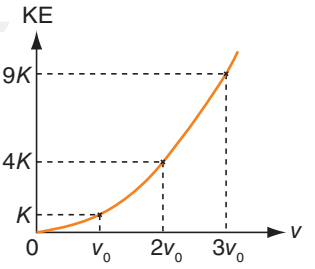


Fig 6.2e *Graph of KE against v for constant m.*

Example 3 | KE of different objects

Find the KE of

(a) a 5-kg bowling ball moving at 7 m s^{-1},

(b) a 4-g bullet moving at 1080 km h^{-1},

(c) a hydrogen molecule of mass 3.35×10^{-27} kg moving at 1930 m s^{-1}.

Solution

Apply $\text{KE} = \frac{1}{2}mv^2$.

(a) $\text{KE} = \frac{1}{2} \times 5 \times 7^2 = 123$ J

(b) $\text{KE} = \frac{1}{2} \times (4 \times 10^{-3}) \times \left(\frac{1080}{3.6}\right)^2 = 180$ J

(c) $\text{KE} = \frac{1}{2} \times (3.35 \times 10^{-27}) \times 1930^2 = 6.24 \times 10^{-21}$ J

Think about...

Is the KE of a fast object always larger than that of a slow object? Why?

Google Forms

Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

1 Several objects each of mass m move at speed v. Complete the table below.

Object	m	v	KE
A	2 kg	3 m s^{-1}	
B	5 g		25 J
C		12 m s^{-1}	3600 J
D	700 kg	90 km h^{-1}	

DSE goal

State that gravitational potential energy is the energy possessed by an object due to its position under gravity.

Derive gravitational PE = mgh.

Solve problems involving gravitational potential energy.

2 Potential energy

There are many types of potential energy (PE). **Elastic potential energy** and **gravitational potential energy** are two of them.

a Elastic potential energy

Elastic potential energy is the energy possessed by an elastic object when the object is stretched, compressed, or bent. The following are two examples (Fig 6.2f and 6.2g).



Fig 6.2f *A rubber band does not possess elastic PE when it is relaxed. It possesses elastic PE when it is stretched.*

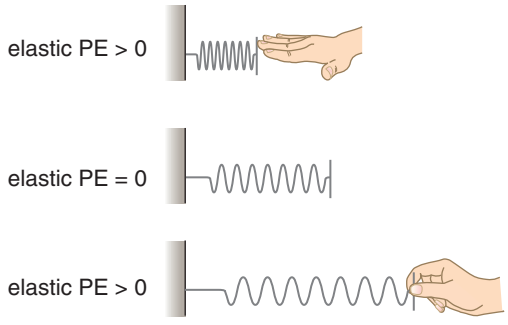


Fig 6.2g *A spring gains elastic PE when it is compressed or extended.*

elastic potential energy 彈性勢能　　gravitational potential energy 重力勢能

b Gravitational potential energy

Gravitational potential energy is the energy possessed by an object due to its position under gravity. When it is raised to a higher position, it gains gravitational PE (Fig 6.2h). You will learn to derive a formula for gravitational PE.



Fig 6.2h *The box gains gravitational PE when it is lifted up to a higher position.*

Simulation

→ This simulation allows Ss to lift a mass up a vertical height and find out how the work done on the mass is related to the change in gravitational potential energy.

To convince Ss who do not feel good with the object starting to move at constant velocity from the ground, Ts may draw two horizontal lines, with the lower one taken as the reference level, on the blackboard and raise a book steadily from a position below the lower line. Then ask Ss to focus on the book when it moves between the two lines.

In Figure 6.2i, an object of mass m is lifted by a force F from the ground at a constant velocity. Since the object does not accelerate, the net force acting on it is zero. Therefore,

$$F = mg$$

Suppose the object is raised for a height h. The work done by F can be written as:

$$W = Fs = mg \times h = mgh$$

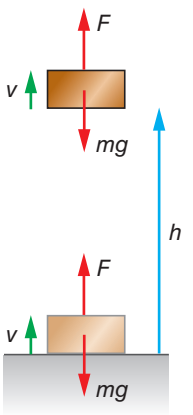


Fig 6.2i *An object is lifted to a height h above the ground.*

The object does not gain KE as its speed does not change. Therefore, all the work done by F appears as an increase in the gravitational PE of the object. If the ground level is taken as the reference level where gravitational PE = 0, the gravitational PE of the object at height h above the ground is given by

gravitational potential **energy** of object (J) ····· **mass** of object (kg)

$$\textbf{Gravitational PE} = mgh$$

acceleration due to gravity ≈ 9.81 m s^{-2} ····· **height** above reference level (m)

Note that:

1 The reference level can be chosen arbitrarily.

2 Gravitational PE is a scalar. It will be negative if the object is below the reference level.

3 Gravitational PE depends on the height (vertical displacement) lifted against gravity, not the distance travelled or path taken (Fig 6.2j).

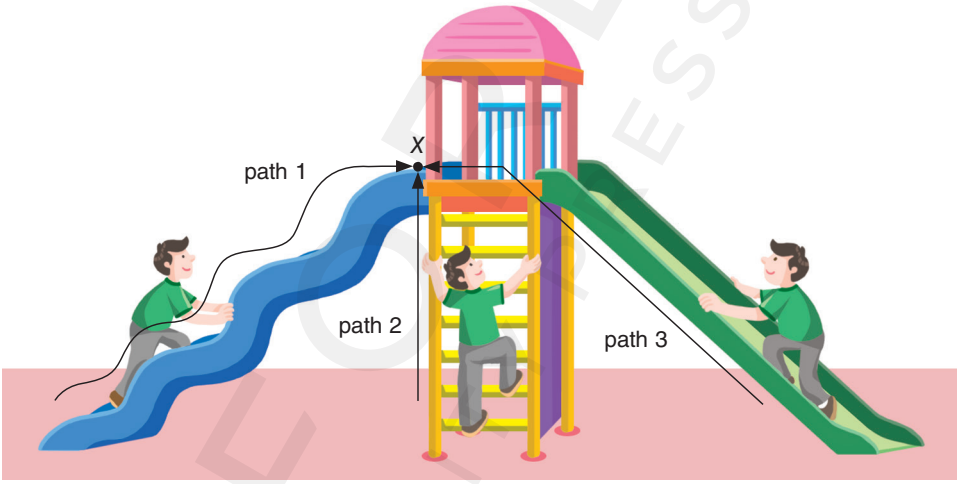


Fig 6.2j *The boy always gains the same gravitational PE, no matter which path he takes to reach X from the ground.*

4 For an object moving from *A* to *B* (Fig 6.2k), its change in gravitational PE is given by

Gravitational PE = $mgh_B - mgh_A$
$= mg(h_B - h_A)$
$= mg\Delta h$

Change in gravitational PE = $mg\Delta h$

change in height (m)

This shows that the change in gravitational PE depends on the change in height (Δh) from one position to the other, but is **independent** of the choice of the reference level.

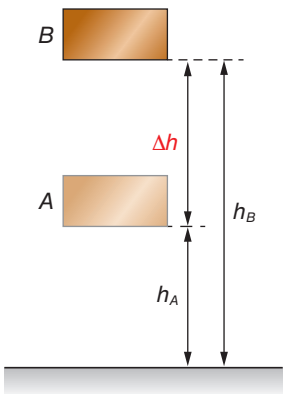
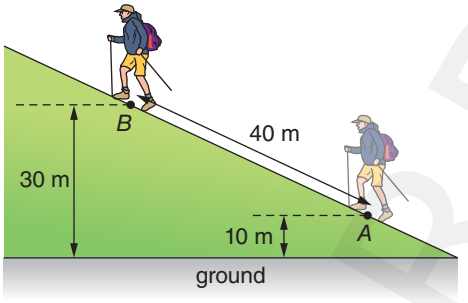


Fig 6.2k *Change in gravitational PE = mgΔh.*

Example 4 PE of a hiker

Johnny of mass 70 kg climbs up a slope. He walks 40 m from point A to point B. A and B are respectively 10 m and 30 m above the ground. Take the gravitational PE at the ground level as zero.



(a) Find Johnny's gravitational PE at A.

(b) What is the change in gravitational PE of Johnny when he walks from A to B?

(c) Now, take the gravitational PE at B as zero. How are the answers to (a) and (b) affected?

Solution

(a) Gravitational PE at $A = mgh_A = 70 \times 9.81 \times 10 = 6870$ J

(b) Change in gravitational PE $= mg\Delta h = 70 \times 9.81 \times (30 - 10) = 13\,700$ J

(c) By taking the gravitational PE at B as zero, $h_A = -20$ m

Gravitational PE at $A = mgh_A = 70 \times 9.81 \times (-20) = -13\,700$ J

$\therefore$ The answer to (a) becomes −13 700 J.

Since Δh is still 20 m, the answer to (b) remains unchanged.

Quick check 3

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Several objects are moved up and down. The change in height is Δh. The mass of each object is m. Complete Table a.

Object	m	Δh	ΔPE
A	20 kg	3 m	
B	0.1 kg		−0.5 J
C		10 cm	30 J

Table a

2 Sue and Rose, having the same mass, travel to The Peak. Sue takes the bus while Rose takes the peak tram (Fig a). Who gains more potential energy?



Fig a

Exercise 6.2

1 Which of the following statements about kinetic energy (KE) is correct?

A The KE of an object can be negative.

B A faster object always has more KE than a slower object.

C The KE of a stationary object must be zero.

D It is a vector quantity.

2 A girl of mass 12 kg goes down a slide as shown (Fig a). Find the loss in PE of the girl.



Fig a

3 Find the total mechanical energy of each of the following objects. Take PE at the ground level as zero.

(a) A bird of mass 0.8 kg flying in the sky at 3 m s^{-1} at a height of 200 m above the ground

(b) A fish of mass 12 kg swimming at 1.8 m s^{-1} at 50 m below ground level

*4 A car of mass m travels with speed u along a horizontal straight road. It then accelerates uniformly for a displacement s and its speed becomes v. The net force acting on the car is F_{net}. By considering the work done by F_{net}, show that the energy that the car gains during travelling is $\frac{1}{2}mv^2 - \frac{1}{2}mu^2$.

*5 Box A is initially at rest while box B initially moves at 1 m s^{-1}. The mass of each box is 2 kg. If both boxes accelerate to 5 m s^{-1}, what is the change in KE of each box?

*6 A lift of mass m moves upwards at a constant velocity for a height Δh from the 2nd floor to the 5th floor (Fig b). The upward force acting on the lift is F. By considering the work done by F, show that the energy that the lift gains during the travel is $mg\Delta h$.



Fig b

*7 A ball accelerates from rest to a speed v and the gain in its KE is ΔE. Find the gain in KE of the ball in each of the following cases.

(a) The ball accelerates from rest to a speed $\frac{v}{2}$.

(b) The ball accelerates from a speed $\frac{v}{2}$ to v.

*8 A ball is thrown vertically upwards from X (Fig c). After reaching Y, it starts to drop and finally hits the ground at Z. X and Y are 1.2 m and 2 m above the ground respectively. The mass of the ball is 0.2 kg. Take PE at the level of X as zero.

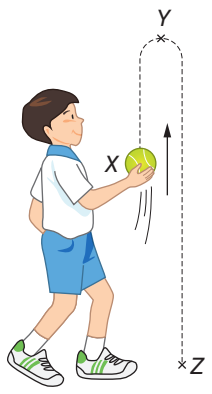


Fig c

(a) At which position is the KE of the ball minimum? Explain.

(b) At which position is the PE of the ball minimum? Explain.

(c) Find the change in PE of the ball when it moves from X to Y.

6.3 Energy conversion and conservation of energy

DSE goal

State the law of conservation of energy.

Discuss the inter-conversion of PE and KE with consideration of energy loss.

Solve problems involving conservation of energy.

DSE exam

13(1B)Q5, 14(1A)Q6, 15(1A)Q8, 16(1A)Q8, 16(1A)Q9, 18(1B)Q3, 20(1B)Q4, 22(1A)Q11, 22(1B)Q4, 23(1A)Q10, etc

Pre-study Energy conversion

In junior science, we learned that energy can be converted from one form to another. The following shows some examples of energy conversion.

Right after the toy is released, the EPE stored in the spring is converted into the KE of the spring toy.



When we rub our hands, the chemical energy of our body is converted into the internal energy of our body, the KE of our hands and the sound energy.



1 Law of conservation of energy

In the last two sections, we learned about work and mechanical energy. Now we will learn how to solve problems by considering energy conversion. This has to make use of an important law about energy—the **law of conservation of energy**:

The total amount of energy in a closed system is conserved, i.e. always kept constant. Energy cannot be created or destroyed.

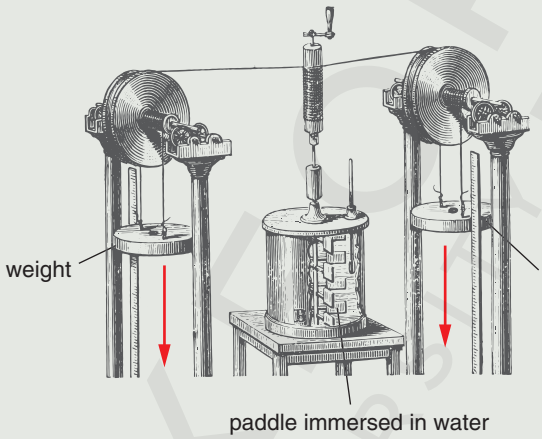
law of conservation of energy 能量守恆定律

We already learned about this law in Book 1. In fact, the law is very useful for calculating energy in different energy conversion processes. In the following part, we will apply the law to the inter-conversion of KE and PE.

Know more

Development of the law of conservation of energy

Energy is a rather recent concept. For hundreds of years, scientists had been looking for the quantity that stays unchanged in all processes. As the understanding of mechanics, heat, electricity and magnetism accumulated in the 19th century, scientists (such as *Joule*, *Helmholtz* and *Lord Kelvin*) gradually established the concept of energy and identified its various forms. The law of conservation of energy was finally accepted around 1847, 150 years after Newton's publication of his laws of motion.



This apparatus was used by Joule to show that the PE lost by the weights in descending was equal to the internal energy gained by the water due to the rotation of the paddle in water.

Joule's apparatus.

2 Examples: total mechanical energy does not change

A simple pendulum is formed by fixing one end of a string and hanging a bob or a mass at the other end. If the bob is pulled to one side and then released, it will swing to and fro (Fig 6.3a). We will study the energy conversion in a simple pendulum in the following experiment.

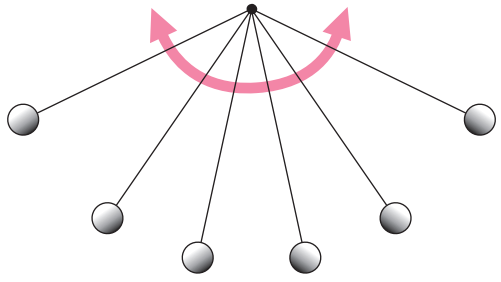


Fig 6.3a *A simple pendulum.*

Joule 焦耳　Helmholtz 亥姆霍茲　Lord Kelvin 開爾文勳爵

Video

→ This video demonstrates Expt 6a.

- Wear safety spectacles.
- Avoid hitting the light gate with the weight.
- Keep a safe distance from the pendulum in motion.

→ This is a Video Quiz about this experiment.

If a sphere is used, its centre may not cut across the light beam of the light-gate. Therefore, the part of the sphere that blocks the light beam is shorter than the diameter of the sphere. This makes the measured time shorter and v larger. ▶

Experiment 6a / Energy changes in a simple pendulum

1 Measure the diameter d and the mass m of the weight.

2 Set up the apparatus as shown (Fig a). The weight should pass through the light-gate when it is at the lowest position.

3 Pull the weight to one side. Measure its height h above its lowest position (Fig b).



Fig a

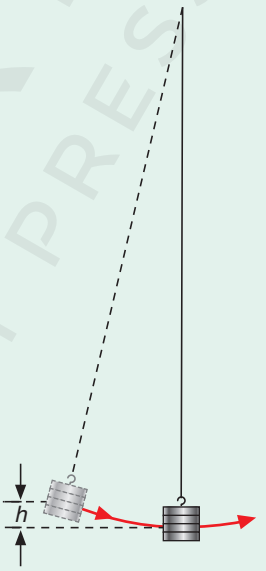


Fig b

4 Start data logging and release the weight.

5 Record the time t taken for the weight to pass through the light-gate.

6 Calculate the speed v of the weight when it passes through the lowest position by $v = \frac{d}{t}$.

7 Calculate the KE of the weight when it is at the lowest position.

8 Calculate the loss in PE when the weight moves from the highest to the lowest position.

Precaution

1 Use a cylindrical weight instead of a sphere to make the pendulum.

2 The weight should pass through the light-gate perpendicularly.

Discussion

1 Is the PE that the weight loses equal to the KE that it gains?

2 Suggest two sources of error.

3 If the weight is replaced by a ping-pong ball, how would the result be affected?

The simple pendulum (weight) can be regarded as a closed system. In the above experiment, we can see that the total mechanical energy of the weight remains the same while the weight is swinging. By considering the weight moving from a position (denoted by 1) to another (denoted by 2),

$$KE_1 + PE_1 = KE_2 + PE_2$$

or loss in KE / PE = gain in PE / KE

These relations can be applied to the energy conversion process in other closed systems, provided that resisting forces such as friction and air resistance are negligible (i.e. the work done against resistive forces can be considered to be zero).

More generally, the gain in one form or some forms of mechanical energy is equal to the loss in other form(s) of mechanical energy.

Simulation

→ This simulation allows Ss to swing a pendulum bob to and fro and find out about the energy change in the process.

Let us study the energy conversion in a simple pendulum in detail. The total mechanical energy in a simple pendulum is conserved if the air resistance acting on the bob is negligible. Figure 6.3b shows the energy conversion in the pendulum as the bob swings to and fro.

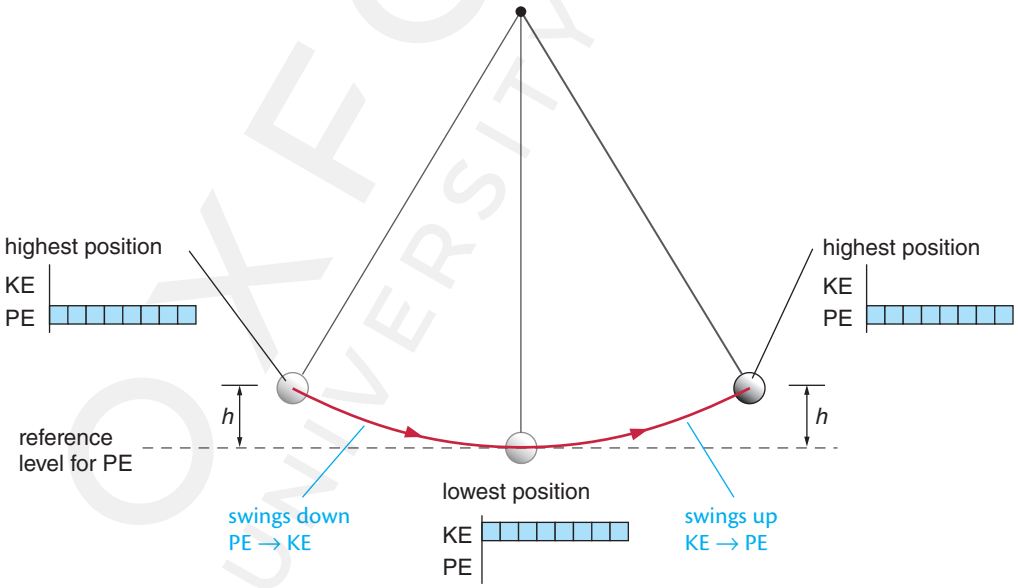


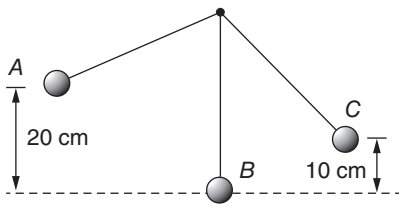
Fig 6.3b *Energy conversion in a simple pendulum, taking PE at the lowest position as zero.*

Example 5 Speed of a pendulum bob

A pendulum bob is pulled to one side and held at *A*, which is 20 cm higher than its lowest position *B*. It is then released.

(a) Find its speed at *B*.

(b) Find its speed at *C* on the other side which is 10 cm above *B*.



Solution

Method 1

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Solving problems with $KE_1 + PE_1 = KE_2 + PE_2$

① Choose a reference level where PE = 0.
② Determine the speeds and heights of the object at the initial and final positions.
③ Set an equation with $KE_1 + PE_1 = KE_2 + PE_2$.
④ Solve the equation.

Take the PE at B as zero.

(a) Consider the motion from A to B.

$KE_A + PE_A = KE_B + PE_B$

$$0 + mgh_A = \frac{1}{2}mv_B^2 + 0$$

$$9.81 \times 0.2 = \frac{1}{2} \times v_B^2$$

$$v_B = 1.98 \text{ m s}^{-1}$$

Its speed at B is 1.98 m s^{-1}.

(b) Consider the motion from A to C.

$KE_A + PE_A = KE_C + PE_C$

$$0 + mgh_A = \frac{1}{2}mv_C^2 + mgh_C$$

$$9.81 \times 0.2 = \frac{1}{2} \times v_C^2 + 9.81 \times 0.1$$

$$v_C = 1.40 \text{ m s}^{-1}$$

Its speed at C is 1.40 m s^{-1}.

Method 2

→ Teaching notes and exercises are available on OUP web.

Skill *Problem-solving*

Solving problems with loss in KE / PE = gain in PE / KE

① Determine the speeds of the object at the initial and final positions and its change in height.
② Determine whether the KE and PE of the object has a gain or a loss.
③ Set an equation with loss in KE / PE = gain in PE / KE.
④ Solve the equation.

(a) Consider the motion from A to B.

Loss in PE = gain in KE

$$mg\Delta h = \frac{1}{2} \times m \times (v^2 - u^2)$$

$$9.81 \times 0.2 = \frac{1}{2} \times (v^2 - 0^2)$$

$$v = 1.98 \text{ m s}^{-1}$$

Its speed at B is 1.98 m s^{-1}.

(b) Consider the motion from A to C.

Loss in PE = gain in KE

$$mg\Delta h = \frac{1}{2} \times m \times (v^2 - 0^2)$$

$$9.81 \times (0.2 - 0.1) = \frac{1}{2} \times v^2$$

$$v = 1.40 \text{ m s}^{-1}$$

Its speed at C is 1.40 m s^{-1}.

Think about...

1 Can Newton's second law be applied to solve this problem? Why?

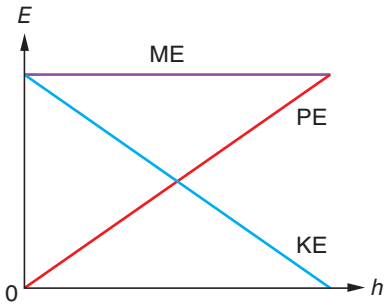
2 Is the following calculation for (a) correct?

$v^2 = u^2 + 2as$

$v^2 = 0^2 + 2 \times 9.81 \times 0.2$

$v = 1.98 \text{ m s}^{-1}$

Figure 6.3c shows how the mechanical energy (ME), PE and KE of the pendulum bob vary with its height h above the lowest position (reference level).



Graph	Reasoning
ME–h graph	ME conserved ∴ A horizontal line
PE–h graph	PE = $mgh \propto h$ ∴ A straight line through the origin with a positive slope (PE reaches maximum when PE = ME, i.e. at the highest position)
KE–h graph	As KE + PE = ME = constant, KE = constant – PE ∴ A straight line obtained by flipping the PE–h graph upside down (KE = ME when PE = 0, KE = 0 when PE = ME)

Fig 6.3c *How ME, PE and KE vary with h.*

Simulation

→ This simulation allows Ss to project a ball up using a compressed spring and find out about the energy changes as the ball rises and falls.

Consider a girl bouncing on a trampoline. The total mechanical energy (KE + gravitational PE + elastic PE) of the system of the girl and the trampoline is conserved if all resistive forces, such as air resistance, are negligible. Figure 6.3d shows the energy conversion in the system.

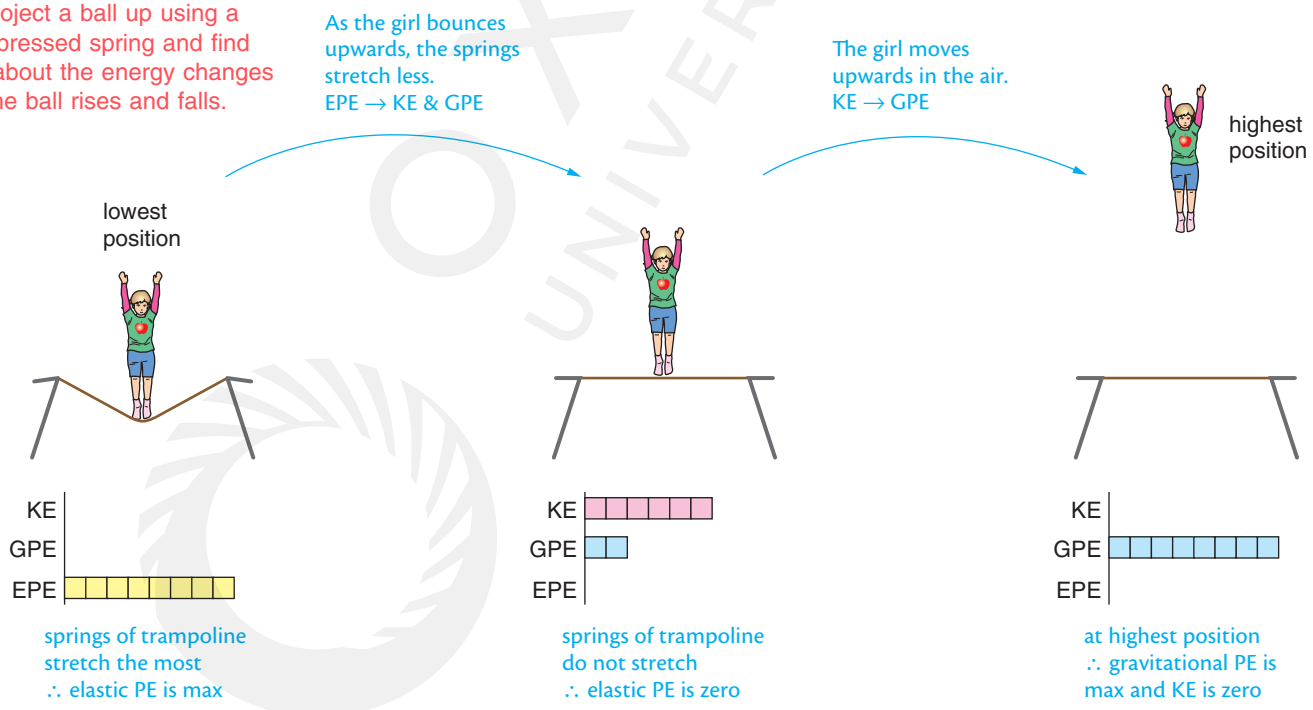
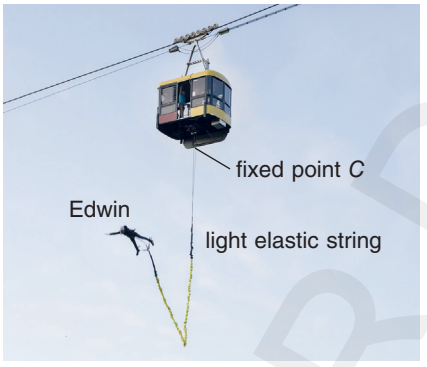
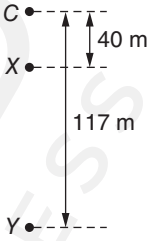


Fig 6.3d *Energy conversion in trampoline jumping, taking gravitational PE at the lowest position as zero.*

The following example is about bungee jumping, which also involves inter-conversion of KE, gravitational PE and elastic PE.

Example 6 Bungee jumping





Edwin goes bungee jumping. He is tied to a fixed point C with a light elastic string and then jumps down from C from rest. The elastic string is 40 m long before stretching. The mass of Edwin is 70 kg. Assume that the total mechanical energy of Edwin and the elastic string remains constant.

Edwin passes points X and Y during his fall.

(a) X is 40 m below C. Find Edwin's speed v_X at X.

(b) Y is 117 m below C and Edwin's speed is 9.95 m s^{-1} at Y. Find the energy EPE_Y stored in the elastic string when Edwin is at Y.

Solution

Take the gravitational PE at Y as zero.

(a) Consider the motion from C to X.

$$\text{KE}_C + \text{EPE}_C + \text{GPE}_C = \text{KE}_X + \text{EPE}_X + \text{GPE}_X$$

$$0 + 0 + mgh_C = \frac{1}{2}mv_X^2 + 0 + mgh_X$$

$$9.81 \times 117 = \frac{1}{2} \times v_X^2 + 9.81 \times (117 - 40)$$

$$v_X = 28.0 \text{ m s}^{-1}$$

Alternatively, the problem can be solved by applying gain in KE = loss in GPE

Tip
EPE = 0 as the string is not taut yet.

(b) Consider the motion from C to Y.

$$\text{KE}_C + \text{EPE}_C + \text{GPE}_C = \text{KE}_Y + \text{EPE}_Y + \text{GPE}_Y$$

$$0 + 0 + mgh_C = \frac{1}{2}mv_Y^2 + \text{EPE}_Y + 0$$

$$70 \times 9.81 \times 117 = \frac{1}{2} \times 70 \times 9.95^2 + \text{EPE}_Y$$

$$\text{EPE}_Y = 76\,900 \text{ J}$$

Alternatively, the problem can be solved by applying loss in GPE = gain in KE + gain in EPE

Know more

Motion in bungee jumping

We can analyse the motion of a person bungee jumping by dividing the motion into several stages.

▸ p.276

Think about...

1 How will the answers change if the GPE at C is taken as zero?

2 Can Newton's second law of motion be applied to solve (a)? Why?

Writing practice

Question

Consider the situation in Example 6 on p.250. Assume that Edwin's speed is maximum when he is at point Z (Fig a). Describe the energy conversion of Edwin from the beginning of the jump to the moment when he reaches point Y. (3 marks)



Fig a

Answer

Gravitational PE is converted to KE from C to X. 1A

Gravitational PE is converted to KE and elastic PE in the string from X to Z. 1A

Gravitational PE and some KE are converted to elastic PE in the string from Z to Y. 1A

Tip
height ↓,
speed ↑,
string not stretched

Tip
height ↓,
speed ↑,
string stretches more and more

Tip
height ↓,
speed ↓,
string stretches even more and more

Practise yourself

1 A pendulum bob is pulled to one side to position A and then released (Fig b). Describe the energy conversion of the bob as it swings from A to C on the other side via the lowest position B. (2 marks)

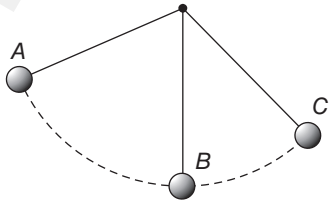


Fig b

2 A girl is bouncing on a trampoline. She falls from rest from position A, starts touching the trampoline at position B and then reaches the lowest position C (Fig c). Describe the energy conversion of the girl-trampoline system as the girl moves from A to C via B. (2 marks)

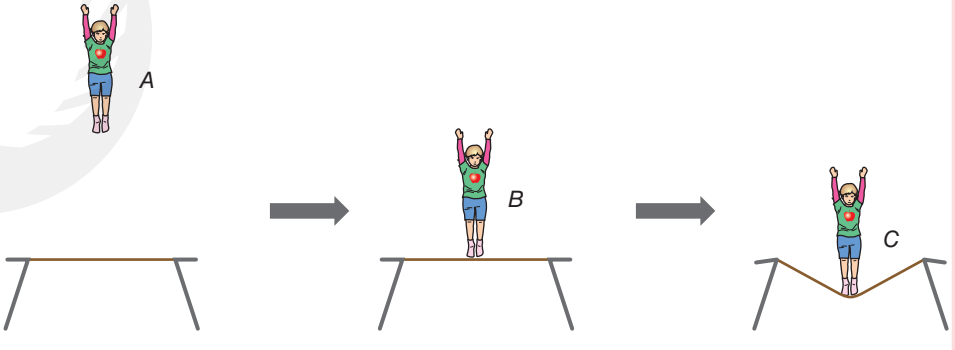


Fig c

Learn by practice 3

Unless otherwise specified, assume air resistance to be negligible.

1 A skier of mass 65 kg moves down from rest at *A* along a frictionless path *ABC* (Fig a). Find the KE of the skier at *B* and *C*.

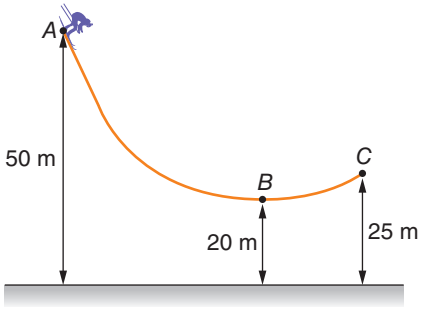


Fig a

2 A pendulum bob of mass 0.5 kg is pulled to one side and held at *P*. It is then released from rest and starts swinging. When it reaches the lowest position *Q*, its speed is 4 m s^{-1}. Given *Q* is at 1.2 m above the ground, find the height of *P* above the ground.

3 A man of mass 60 kg jumps on a trampoline. His centre of gravity (c.g.) is 2.5 m above the ground when reaching the highest position. Then he falls down onto the trampoline and his c.g. is 1.5 m above the ground when reaching the lowest position. Find the elastic PE stored in the trampoline when he reaches the lowest position. Assume that the total mechanical energy of the man and the trampoline remains constant.

3 Examples: total mechanical energy changes

You must know that a bicycle moving on a horizontal road will slow down if you do not pedal. Can you explain this phenomenon in terms of energy?

The total mechanical energy of a system is conserved if **the resistive forces acting on the system are negligible**. However, when the bicycle moves, it experiences strong friction and air resistance. Work is done against these forces. The bicycle's mechanical energy is lost as other forms of energy, such as internal energy (Fig 6.3e). As a result, the rider has to do work (pedalling) to maintain the bicycle's speed.

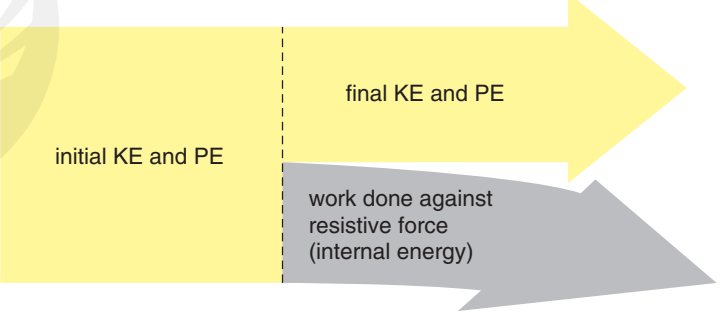


Fig 6.3e *A Sankey diagram showing the situation that some mechanical energy is lost as other forms of energy due to the work done.*

Simulation

→ This simulation shows the energy changes when a block slides down an inclined plane. The height and the friction of the inclined plane can be varied.

Mathematically, the energy conversion can be expressed as follows.

$$\underbrace{KE_1 + PE_1}_{\text{initial mechanical energy}} = \underbrace{KE_2 + PE_2}_{\text{final mechanical energy}} + \text{work done against resistive force}$$

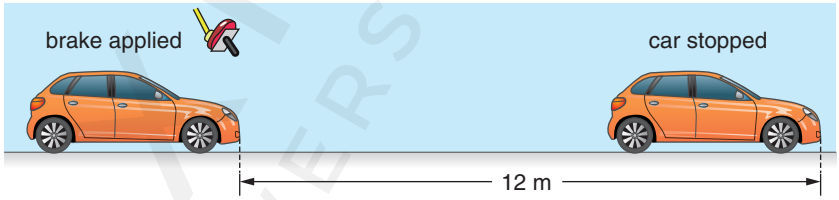
or $\text{loss in KE / PE} = \text{gain in PE / KE} + \text{work done against resistive force}$

The work done against resistive force is on the 'gain side' as it is equal to the gain in some forms of non-mechanical energy.

Note that the law of conservation of energy still applies here. Although there is a loss in the total mechanical energy, the total amount of all forms of energy are conserved.

Example 7 Energy conversion in car braking

A car travels on a horizontal road at a speed of 18 m s^{-1}. After the driver applies the brake, the car is brought to a stop in 12 m (stopping distance). Take the mass of the car to be 900 kg. Assume that the friction acting on the car during braking is constant.



(a) Estimate the friction acting on the car.

(b) If the road is wet, the friction acting on the car will be smaller. How will the stopping distance change? Explain briefly.

Solution

Alternatively, the problem can be solved by applying $KE_1 = KE_2 + W$ against f

(a) Loss in KE = W against friction f

$$\frac{1}{2}m(u^2 - v^2) = fs$$

$$\frac{1}{2} \times 900 \times (18^2 - 0) = f \times 12$$

$$f = 12\,200 \text{ N}$$

The friction acting on the car is 12 200 N.

(b) The work done against friction W needed to stop the car will remain unchanged (equal to initial KE of the car).

By $W = fs$, the stopping distance s will increase if friction f decreases.

Think about...

Can Newton's second law of motion be applied to solve this problem? Why?

Example 8 Work done by tension and elastic energy

Edwin goes bungee jumping again. He is tied to point C with a light elastic string. He jumps down from C from rest, passes point X and then point Y (Fig a). When Edwin is at X, his speed is 28.0 m s^{-1} and the elastic string starts to stretch. Edwin's speed reaches 9.95 m s^{-1} at Y, which is 77 m below X. The mass of Edwin is 70 kg.

Find the average tension T acting on Edwin by the string during his fall from X to Y.

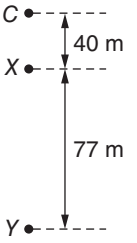


Fig a

Solution

Take the gravitational PE at Y as zero.

Consider the total mechanical energy of Edwin.

$$\text{KE}_X + \text{PE}_X = \text{KE}_Y + \text{PE}_Y + W \text{ against tension } T$$

$$\frac{1}{2}mv_X^2 + mgh_X = \frac{1}{2}mv_Y^2 + 0 + Ts$$

$$\frac{1}{2} \times 70 \times 28.0^2 + 70 \times 9.81 \times 77 = \frac{1}{2} \times 70 \times 9.95^2 + T \times 77$$

$$T = 998 \text{ N}$$

The average tension T acting on Edwin by the string is 998 N.

Tip
As only the mechanical energy of Edwin is considered, the elastic PE of the string is not included in the equation.

Alternatively, the problem can be solved by applying loss in KE + loss in GPE = W against T

Google Forms Microsoft Forms

Quick check 4

→ E-version in OUP Exercise Platform is also available.

1 An object moves along a rough path from points A to B, with B at a level higher than A (Fig a).

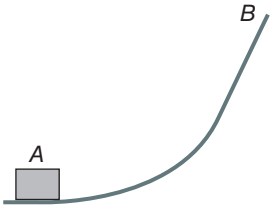


Fig a

(a) Its total mechanical energy at A is (larger than / smaller than / equal to) its total mechanical energy at B.

(b) Its loss in KE is (larger than / smaller than / equal to) its gain in PE.

Learn by practice 4

Unless otherwise specified, assume air resistance to be negligible.

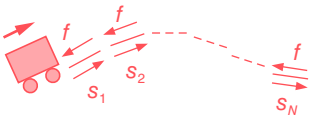
1 A car of mass 1500 kg decelerates uniformly from 20 m s^{-1} to a stop in 30 m along a horizontal road. Find the friction acting on it.

2 A roller coaster cart of mass 500 kg moves along a curved path from *A* to *B* (Fig a). It travels a total distance of 60 m and the friction acting on it is 15 N. Given the speed of the cart is 2.5 m s^{-1} at *A*, find its speed at *B*.



Fig a

By chopping a curved path of length *s* into many short straight segments, the work done against friction along the path, *W*, can be estimated by the sum of the work done against friction along each segment.



In some processes, the total mechanical energy of a system increases. For example, in Figure 6.3f, a force pulls a box to accelerate on a horizontal surface. Positive work is done by a force on the box. As a result, some other forms of energy are converted into the box's mechanical energy, causing its total mechanical energy to increase (Fig 6.3g).

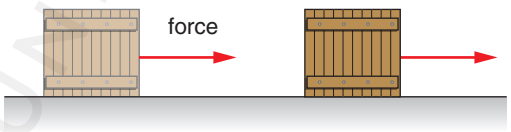


Fig 6.3f *A pulling force does positive work on a box.*

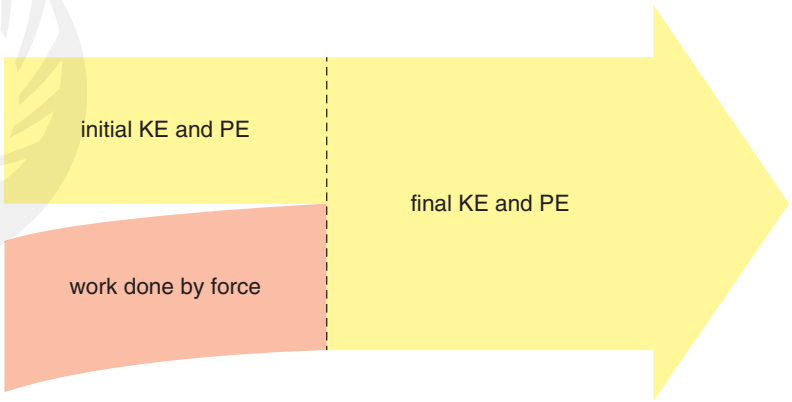


Fig 6.3g *A Sankey diagram showing the situation that the total mechanical energy increases due to the work done.*

Mathematically, the energy conversion can be expressed as follows.

$$\underbrace{KE_1 + PE_1}_{\text{initial mechanical energy}} + \text{work done by force} = \underbrace{KE_2 + PE_2}_{\text{final mechanical energy}}$$

or

$$\text{loss in KE / PE} + \text{work done by force} = \text{gain in PE / KE}$$

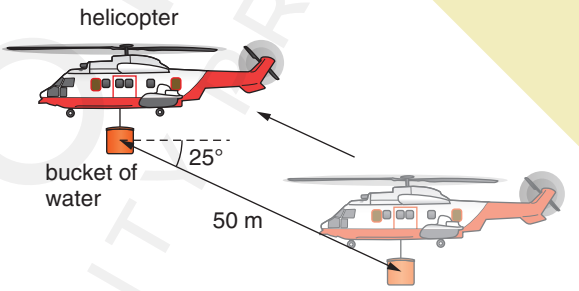
☆ HKDSE 2016 Paper 1A Q9

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Work done and change in PE

A helicopter carrying a bucket of water travels 50 m at a constant velocity as shown. The mass of the bucket of water is 2000 kg. What is the work done on the bucket of water by the helicopter? Assume that air resistance is negligible.



A 0
B 415 000 J
C 889 000 J
D 981 000 J

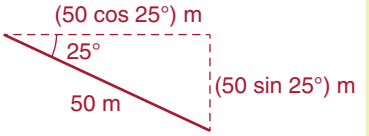
Solution

The KE of the bucket does not change as its speed does not change. Also, there is no resistive force acting on the bucket. Therefore,

work done by helicopter = gain in PE
= $mg\Delta h$
= $2000 \times 9.81 \times 50 \sin 25°$
= 415 000 J

Common mistake ⚠
Students may not notice this relationship.

Common mistake ⚠
Students may wrongly think that $W = 2000 \times 9.81 \times 50 \cos 25°$

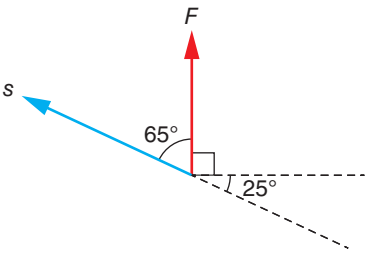


∴ The answer is B.

Alternative solution:

Since the bucket travels at a constant velocity, $F_{net} = 0$.

As there is no resistive force acting on the bucket, the force F acting on the bucket by the helicopter is equal to the bucket's weight mg in magnitude and points upwards.



Work done by helicopter = $Fs \cos \theta$
= $mg \times 50 \times \cos 65°$
= $2000 \times 9.81 \times 50 \times \cos 65°$
= 415 000 J

∴ The answer is B.

▶ Revision 6 Q18 (p.270)

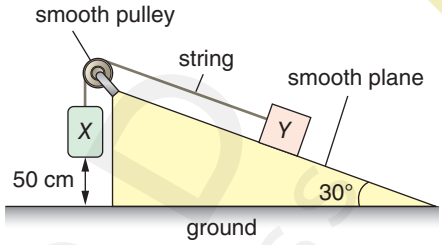
☆ HKDSE 2019 Paper 1A Q12

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Motion of two connected objects

Block Y of mass 1 kg is placed on a smooth plane which inclines at 30° to the horizontal. It is connected to block X of mass 2 kg by a string over a smooth pulley. The blocks are released from rest when X is 50 cm above the ground as shown. What is the speed of X just before hitting the ground?



A $1.81\ \text{m s}^{-1}$ B $1.93\ \text{m s}^{-1}$

C $2.21\ \text{m s}^{-1}$ D $3.13\ \text{m s}^{-1}$

Solution

PE lost by X = PE gained by Y + KE gained by X and Y

$$m_X g\Delta h_X = m_Y g\Delta h_Y + \frac{1}{2}(m_X + m_Y)(v^2 - u^2)$$

$$2 \times 9.81 \times 0.5 = 1 \times 9.81 \times (0.5 \sin 30°) + \frac{1}{2} \times (2 + 1) \times (v^2 - 0^2)$$

$$v = 2.21\ \text{m s}^{-1}$$

∴ The answer is C.

Common mistake ⚠
Students may wrongly consider the motion of X as free fall and choose D.

Common mistake ⚠
Students may have difficulty in finding Δh_Y.

Alternative solution (using Newton's second law):

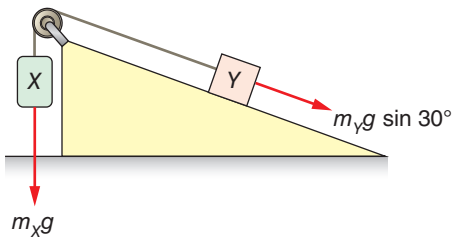
Consider the system of blocks X and Y.

By $F_{net} = ma$,

$$m_X g - m_Y g \sin 30° = (m_X + m_Y)a$$

$$2 \times 9.81 - 1 \times 9.81 \times \sin 30° = (2 + 1) \times a$$

$$a = 4.905\ \text{m s}^{-2}$$



Consider X. Take downwards as positive.

$s = +0.5$ m, $u = 0$, $a = +4.905\ \text{m s}^{-2}$, $v = ?$

By $v^2 = u^2 + 2as$,

$v^2 = 0^2 + 2 \times 4.905 \times 0.5$

$v = 2.21\ \text{m s}^{-1}$

∴ The answer is C.

▶ Revision 4 Q19 (p.187)

You may see that Newton's laws of motion are applicable to some of the questions in this chapter. In general, when the law of conservation of energy and Newton's laws of motion are both applicable, the energy approach often requires fewer steps. However, if you need to find out acceleration or time, Newton's laws and equations of motion should be used.

Exercise 6.3

Unless other specified, assume air resistance to be negligible.

1 In a water park, a boy slides down along a smooth water chute as shown (Fig a). A is a point at 10 m above the water level and B is a point at the water level. Assume that the resistive force acting on the boy is negligible.

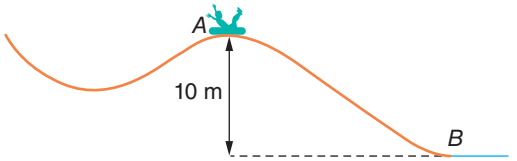


Fig a

If the speed of the boy at A is 3.4 m s^{-1}, what is his speed when he reaches B?

A 10.6 m s^{-1}

B 14 m s^{-1}

C 14.4 m s^{-1}

D Cannot be determined as the mass of the boy is not given.

***2** A pendulum bob of mass 0.5 kg is released from rest at X (Fig b). Take the PE at Y as zero.

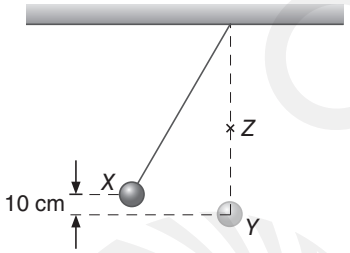


Fig b

Which of the following statements is/are correct?

(1) The mechanical energy of the bob is 0.491 J.

(2) If the mass of the bob is larger, its speed at Y will be higher.

(3) If a pin is fixed at Z, the maximum height that the bob reaches will not be at the same level as X.

A (1) only

B (3) only

C (1) and (2) only

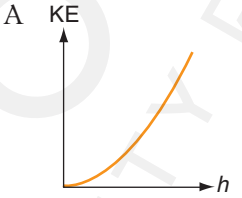
D (1), (2) and (3)

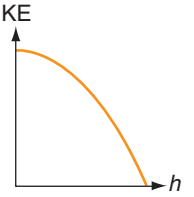
***3** A skateboarder slides down a ramp from rest (Fig c).

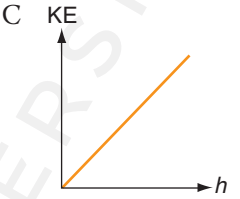


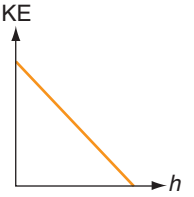
Fig c

Which of the following graphs best shows how his KE changes with his height h above the lowest point of the ramp? Assume that the friction acting on him is negligible.

A


B


C


D


***4** Solve the following problems using (i) energy approach, and (ii) Newton's laws of motion and the equations of motion.

(a) A stone is thrown vertically downwards from a cliff with an initial speed of 2 m s^{-1}. Find its speed after it falls for a distance of 5 m.

(b) A box of mass 500 g is released from A on a rough inclined plane (Fig d). Find the friction acting on the box if the box moves with a speed of 3 m s^{-1} at B.

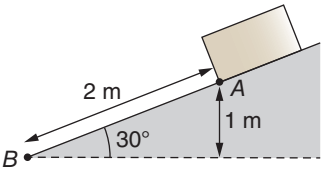


Fig d

(c) A 5-kg box is initially at rest on a rough horizontal surface. Peter pushes it to the right for 3 m by a horizontal force of 20 N. The friction acting on the box is 5 N. Find the final speed of the box.

★**5** A ball is thrown vertically upwards at A with an initial speed of 7 m s^{-1}. B is the highest point that it reaches (Fig e).

(a) How high is B above A?

(b) What is the speed of the ball when it returns to A?

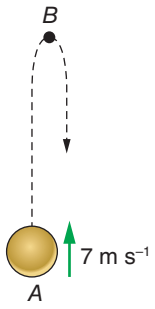


Fig e

★**6** A bullet of mass 50 g penetrates a wall which is 0.15 m thick (Fig f). If the speed of the bullet is 80 m s^{-1} when it enters the wall and is 50 m s^{-1} when it leaves the wall, what is the average resistive force acting on the bullet by the wall?

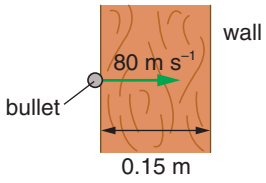


Fig f

★**7** A diver of mass 65 kg stands at A which is 10 m above the water surface (Fig g). She jumps upwards with a speed of 2.5 m s^{-1}. Then she enters the water and reaches the lowest position B.

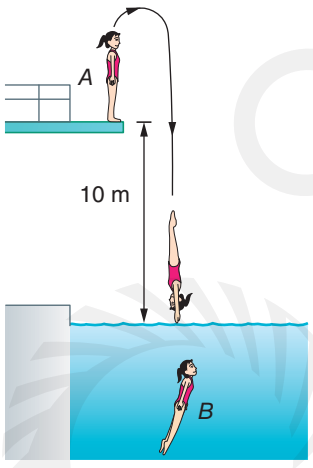


Fig g

(a) If the average resistive force acting on the diver in the water is 2800 N, how far is B below the water surface?

(b) If she falls from A from rest, will she reach B? Explain briefly.

(c) Suggest one form of energy that the mechanical energy of the diver is converted to during the process.

★**8** Metal balls P and Q have the same mass. As shown in Figure h, P is released from rest at level X on a smooth inclined plane. Its speed is v_P when it reaches level Y. Q is hung by a string. It is pulled to level X and then released from rest. Its speed is v_Q when it reaches level Y.

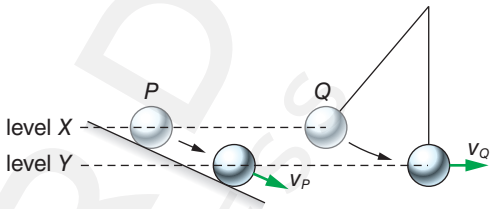


Fig h

Which one is larger, v_P or v_Q? Explain your answer.

★**9** A marble of mass 50 g is projected vertically upwards by a compressed spring (Fig i). The spring is released at time $t = 0$. At $t = t_1$, the elastic PE stored in the spring just becomes zero and the speed of the marble is 2 m s^{-1}. The marble reaches the maximum height at $t = t_2$. The total mechanical energy of the spring and the marble is conserved.

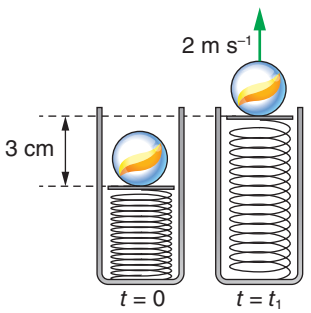


Fig i

(a) Find the maximum height (from the initial level) reached by the marble.

(b) Find the elastic PE stored in the spring at $t = 0$.

(c) Find the magnitude of the average force acting on the marble by the spring from $t = 0$ to t_1.

(d) Describe the energy conversion of the marble and the spring from $t = 0$ to $t = t_2$.

6.4 Power

1 Definition of power

DSE goal

Define power as the rate of energy transfer.

Apply $P = \frac{W}{t}$ to solve problems.

DSE exam

12(1A)Q9, 13(1B)Q3, 14(1B)Q3(d), 17(1A)Q11, 23(1B)Q3(b)

You consume about 5000 J of energy in each of the following activities. But since you spend different lengths of time for the same energy consumption, you consume energy at different rates (Fig 6.4a). In this part, we will see how to describe the rate in a scientific manner.



Fig 6.4a *Consuming the same amount of energy in different periods of time.*

In Book 1 Chapter 3, we learned that **power** is the **rate of energy transfer**, i.e. $P = \frac{Q}{t}$. Similarly, if work W is done in time t, power P can be defined as

1 W = 1 J s^{-1} ▶

$$P = \frac{W}{t}$$

power (W); work done (J); time taken (s)

It tells us how fast work is done or how fast energy is transferred. Figure 6.4b shows some examples of power output.

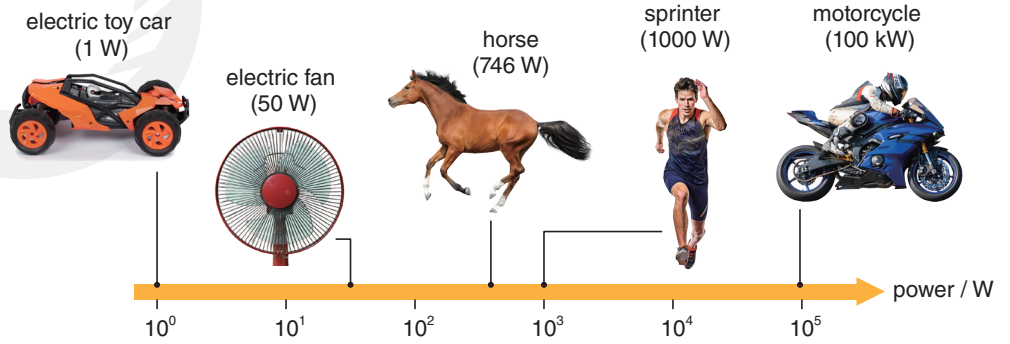


Fig 6.4b *Some examples of power output.*

power 功率

Example 9 Average power

A woman does 5000 J of work in jogging for 10 s.



(a) Find her average power.

(b) A man also does 5000 J of work in walking for 40 s. Is his power larger or smaller than the woman's power? Explain without calculation.

Solution

(a) $P = \frac{W}{t} = \frac{5000}{10} = 500$ W

(b) The man takes more time t to do the same work W. According to $P = \frac{W}{t}$, his power P is smaller.

2 Power and velocity

Suppose a force F acts on an object for time t. The displacement of the object is s in this period (Fig 6.4c). F and s are in the same direction. The power P developed by F is

$$P = \frac{W}{t} = \frac{F \times s}{t} = F \times \frac{s}{t}$$

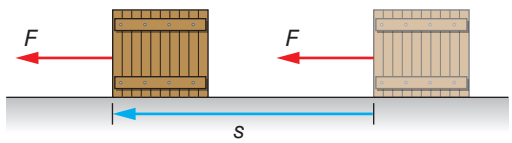


Fig 6.4c *A force F acts on an object for a displacement s within time t.*

If the time t is extremely short, $\frac{s}{t}$ is equal to the instantaneous velocity v. Therefore, the instantaneous power developed by F is

$$P = Fv$$

instantaneous **power** (W)

instantaneous **velocity** (m s^{-1})

force (N)

Assume that F acting on the object is constant (Fig 6.4d).

- If the object speeds up in the process, by $P = Fv$, the power developed by F will increase. In general, the power developed by a certain constant force is not always the same, but changes with the velocity of the object it acts on.
- If there is a resistive force acting on the object so that the object moves with a constant velocity, the object has no change in mechanical energy. The power developed by F is constant and equals the power loss due to the resistive force.

If the object speeds up, $P\uparrow$.
If the object moves with a constant v, P remains unchanged.

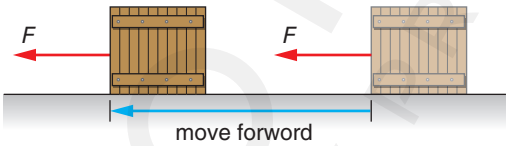


Fig 6.4d *A constant force F acts on an object moving forwards.*

Alternatively, take the moving direction of the car as positive,
power of friction
$= P_f = -800 \times 30 = -24\,000$ W
Since the car travels at a constant velocity on a level road,
net power on car = 0
$P_c + P_f = 0$
$P_c = 24\,000$ W

Tip

Since the car is travelling at a constant velocity, the net force is zero.

Example 10 Power of a car

A car is travelling at a constant velocity of 30 m s^{-1} on a level road. What is the power output of the car engine if the total resistive force acting on the car is 800 N?

Solution

Forward force due to the engine = total resistive force = 800 N

Power output = $Fv = 800 \times 30 = 24\,000$ W

Think about...

Tom claims that if the car is travelling at a higher velocity, a smaller force is needed to keep the velocity constant according to $P = Fv$. Is he correct?

Google Forms Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 A girl does 5000 J of work in swimming for 20 s. Find her average power.

2 A car is travelling on a bridge. Its engine provides a forward force of 1200 N. Find the power output of the engine at a moment when the car's speed is 20 m s^{-1}.

3 *True or false*: If the forward force provided by the engine in Q2 remains unchanged, the power output of the engine must stay the same. (T / F)

4 The power output of an engine is 400 W. How much work does the engine do in 20 s?

Example 11 | Power and conservation of energy I

A spaceship of mass 120 000 kg flies forwards in outer space. Its rocket is fired to speed up the spaceship from 500 m s^{-1} to 600 m s^{-1} in 20 s. Assume that the power provided by the rocket is constant and the resistive force acting on the spaceship is negligible. Find the power of the rocket.

Solution

KE_1 + work done by rocket = KE_2

$$\frac{1}{2}mu^2 + Pt = \frac{1}{2}mv^2$$

$$P = \frac{\frac{1}{2}m(v^2 - u^2)}{t}$$

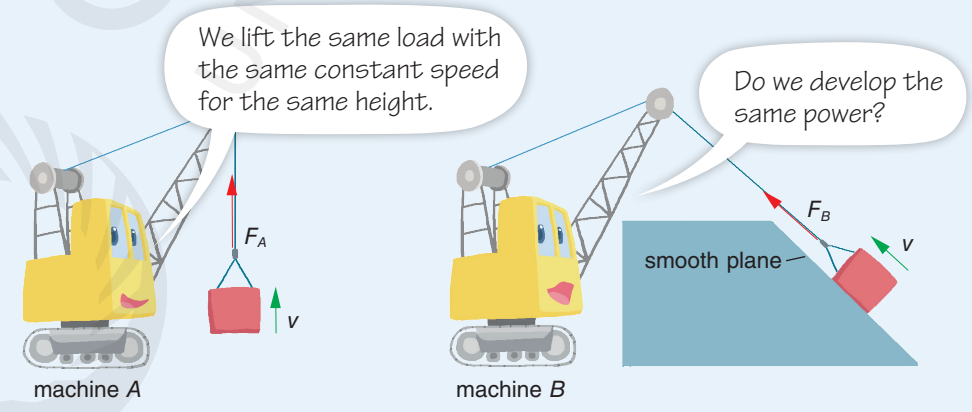
$$P = \frac{\frac{1}{2} \times 120\,000 \times (600^2 - 500^2)}{20}$$

$$= 3.3 \times 10^8 \text{ W}$$

The power of the rocket is 3.3×10^8 W.

Right and wrong

Machines *A* and *B* each lift a load as shown.



✗ Power by *A* = power by *B*

✓ Since $F_A > F_B$, power by *A* > power by *B*

Alternatively, *A* raises the load by a larger height in each second than *B* does.

By $P = \frac{W}{t}$, the power of *A* is higher.

Example 12 | Power and conservation of energy II

A boat of mass 2500 kg moves along a straight line at a constant acceleration of 1 m s^{-2}. At time $t = 0$, its velocity is 2 m s^{-1}. The average power output of its engine is 12 kW from $t = 0$ to $t = 4$ s. Assume the resistive force acting on the boat is constant. By considering the energy conversion of the boat, find the magnitude of the resistive force acting on the boat.



Solution

KE_1 + work done by engine = KE_2 + work done against resistive force f

$$\frac{1}{2}mu^2 + Pt = \frac{1}{2}mv^2 + fs \quad \text{..... (1)}$$

Tip
v, f and s are unknowns.

Take forwards as positive.

$u = +2$ m s^{-1}, $a = +1$ m s^{-2}, $t = 4$ s, $v = ?$, $s = ?$

$$v = u + at = 2 + 1 \times 4$$
$$= 6 \text{ m s}^{-1}$$

$$s = ut + \frac{1}{2}at^2 = 2 \times 4 + \frac{1}{2} \times 1 \times 4^2$$
$$= 16 \text{ m}$$

Substitute into (1):

$$\frac{1}{2} \times 2500 \times 2^2 + 12\,000 \times 4 = \frac{1}{2} \times 2500 \times 6^2 + f \times 16$$

$$f = 500 \text{ N}$$

The magnitude of the resistive force is 500 N.

Think about...

1 Why is the instantaneous power output of the boat engine at $t = 4$ s larger than 12 kW?

2 Find the instantaneous power output of the boat engine at $t = 4$ s.

Exercise 6.4

Unless otherwise specified, assume air resistance to be negligible.

1 In the story *Jack and the Beanstalk*, Jack climbs up a gigantic beanstalk which is 2000 m tall and arrives a land high up in the clouds (Fig a). Assume his mass is 45 kg and his average power is 20 W. How long does Jack take to climb up the beanstalk?

A 4500 s

B 12 300 s

C 44 100 s

D 90 000 s



Fig a

(For Q2–3.) A motor pulls a block along a horizontal floor with a constant force of 500 N (Fig b).

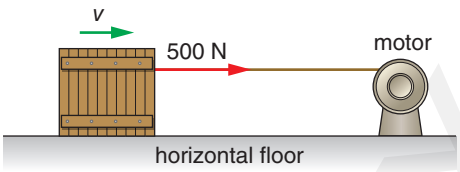


Fig b

2 Suppose the floor is rough and the block moves at a constant velocity. If the power output of the motor is 1000 W, what is the velocity of the block?

A 0.5 m s^{-1}

B 2 m s^{-1}

C 25 m s^{-1}

D 50 m s^{-1}

***3** Suppose the floor is smooth and the block starts moving from rest. Figure c shows the variation of the power output P of the motor with the velocity v of the block. What is P_1?

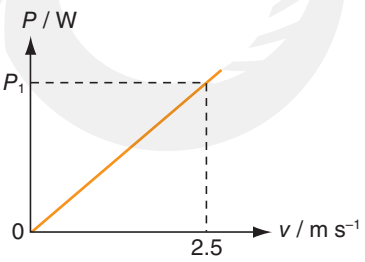
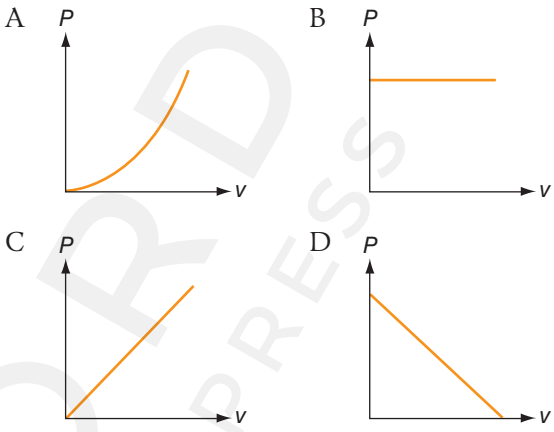


Fig c

A 200 W B 500 W

C 1000 W D 1250 W

***4** A motorcycle starts from rest and accelerates uniformly. Assume that the resistive force acting on it remains unchanged. Which of the following graphs best shows how the power output P of the engine varies with the speed v of the motorcycle?



***5** A block is pulled by a force and accelerates from rest to 1 m s^{-1} on a rough horizontal surface in 4 s. The average power delivered by the force is 8 W. If the mass of the block is 5 kg, what is the work done against friction within this period of time?

***6** A lift transports several people from the ground floor to the 20th floor, which is 60 m high, in 20 s. The total mass of the lift cage and the people inside is 700 kg.

(a) Suppose the lift is raised directly by a motor. Estimate the average power output of the motor.

(b) Suppose the lift is connected to a 500-kg counterweight by a cable over a wheel (Fig d). When the motor raises the lift cage, the counterweight moves downwards. The PE lost by the counterweight is converted into the PE gained by the lift cage. Estimate the average power output of the motor.

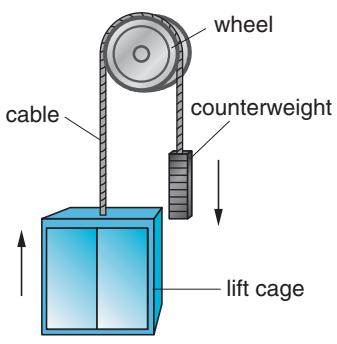
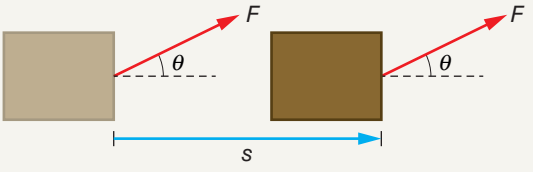


Fig d

Key concepts 6

1 Work done (*W*)

Energy transfer due to a force



$$W = Fs\cos\theta$$

2 Kinetic energy (KE)

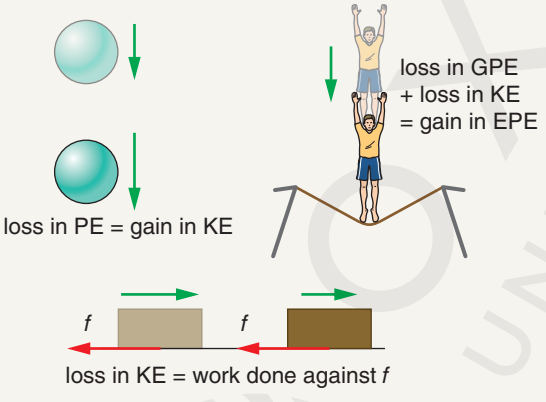
Energy of moving object



$$\text{KE} = \frac{1}{2}mv^2$$

3 Conservation of energy

Energy can be changed from one form into another, but it cannot be created or destroyed.



Elastic potential energy (EPE)

Energy of elastic object when it is

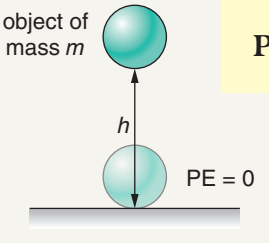
1 stretched 

2 compressed 

3 bent 

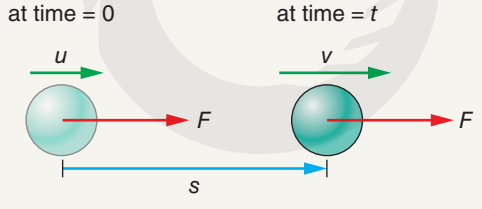
Gravitational potential energy (GPE / PE)

Energy due to the position of object under gravity



$$\text{PE} = mgh$$

4 Power (*P*)



$$\text{Average power } P = \frac{W}{t}$$

$$\text{Instantaneous power } P = Fv$$

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Key terms

1 elastic potential energy 彈性勢能 p.239
2 gravitational potential energy 重力勢能 p.239
3 law of conservation of energy 能量守恆定律 p.244
4 mechanical energy 機械能 p.237
5 mechanical work 機械功 p.230
6 power 功率 p.260
7 work 功 p.230

Revision 6

Multiple-choice questions

Unless otherwise specified, assume air resistance to be negligible.

★1 6.4 A polar bear of mass 500 kg swims 50 m in 40 s at a uniform speed (Fig a). What is the water resistance acting on it if its power output is 200 W?



Fig a

A 160 N B 200 N
C 250 N D 500 N

★2 6.3 A man goes bungee jumping. Which of the following statements are correct when he reaches the lowest position? Assume that he only moves in a vertical direction.

(1) No net force acts on him.
(2) The elastic PE stored in the string is the maximum.
(3) His KE is zero.

A (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

★3 6.3 An object of mass 2 kg moves on a smooth horizontal surface. At time $t = 0$, David starts exerting a force F towards the left on the object. Figure b shows the velocities of the object at $t = 0$, t_1 and t_2.

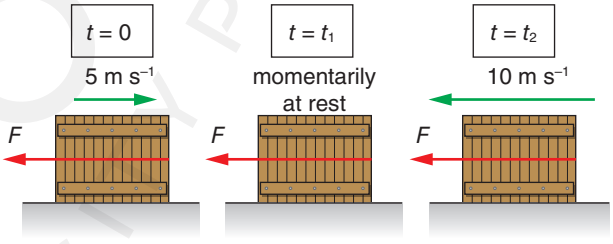


Fig b

What is the total work done on the block by David from $t = 0$ to $t = t_2$?

A 25 J B 75 J
C 125 J D 225 J

★4 6.1 A helium balloon moves along an inclined ceiling as shown (Fig c). The upward force acting on the balloon is 1.2 N.

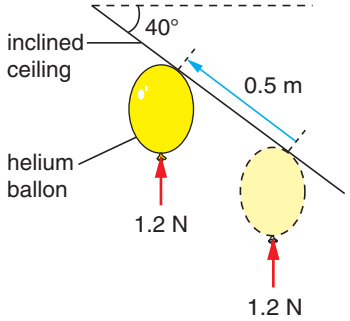


Fig c

What is the work done by the upward force when the balloon has moved 0.5 m along the ceiling?

A 0.386 J
B 0.460 J
C 0.600 J
D 0.715 J

***5** 6.3 An object of mass 3 kg is released from rest from a height 1.5 m above the ground. It passes through a fixed board of 2 cm thick during its fall (Fig d). Its speed is 5 m s^{-1} just before reaching the ground. Find the average resistive force acting on the object when it passes through the board.

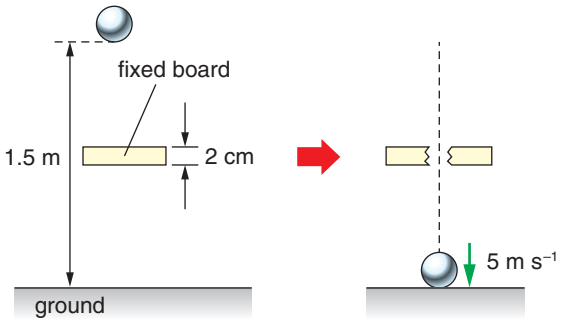


Fig d

A 332 N B 1875 N
C 2210 N D 4080 N

***6** 6.2 A policeman is chasing a thief. The mass of the thief is double that of the policeman and they have the same KE. What is the ratio of the speed of the policeman to that of the thief?

A $1 : \sqrt{2}$ B $1 : 2$
C $\sqrt{2} : 1$ D $2 : 1$

***7** 6.3 A block is allowed to slide down a smooth inclined plane from rest from different positions (Fig e). When the block is released from X, its velocity at Z is v. When the block is released from Y, its velocity at Z is $2v$. If the vertical distance between X and Z is h, what is the vertical distance between Y and Z?

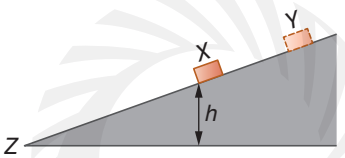


Fig e

A $\sqrt{2}h$ B $2h$
C $4h$ D $8h$

***8** 6.4 A helicopter with a mass of 3000 kg is moving vertically upward at a constant velocity of 10 m s^{-1}. The air resistance acting on the helicopter is 1000 N. What is the average power of the helicopter?

A 5 kW B 10 kW
C 294 kW D 304 kW

***9** 6.3 Lisa of mass 55 kg jumps on a trampoline (Fig f).

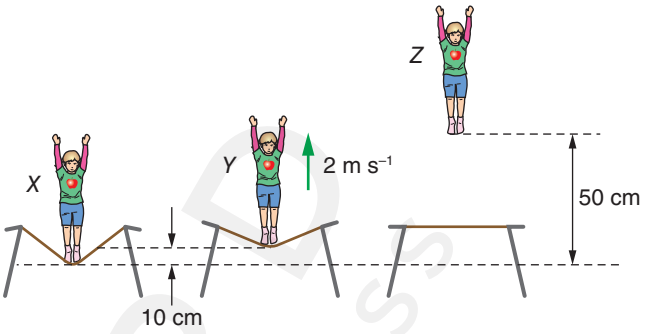


Fig f

At position Y, she is 10 cm above the lowest position X and her speed is 2 m s^{-1}. The highest position Z that she reaches is 50 cm above X. Find the elastic PE stored in the trampoline when she is at Y. Assume that the total mechanical energy of Lisa and the trampoline is conserved.

A 106 J B 160 J
C 216 J D 21 500 J

***10** 6.1 A 20-kg parcel is put on an inclined conveyor belt as shown (Fig g). As the conveyor belt moves, it takes the parcel from X to Y at a constant speed of 2 m s^{-1}.

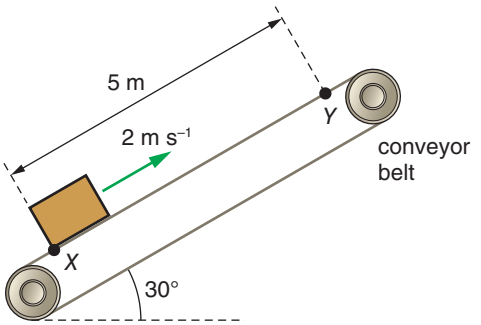
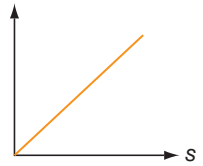


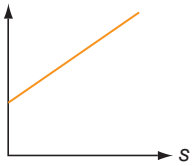
Fig g

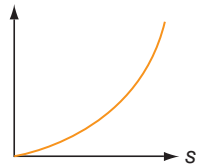
What is the work done by friction on the parcel?

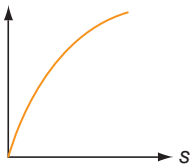
A −491 J B −196 J
C 0 D 491 J

****11** Misc A car accelerates uniformly from rest along a horizontal straight road. The total resistive force acting on the car remains unchanged. Which of the following graphs best describes how the power output P of the car varies with the displacement s during the acceleration?

A 

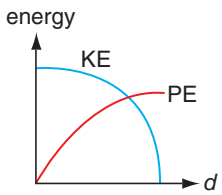
B 

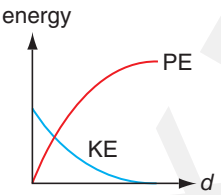
C 

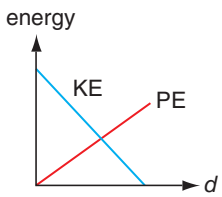
D 

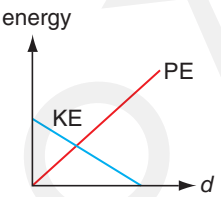
Refer p.63, 261

12 Misc A block is given a sharp push up a rough inclined plane. The friction acting on the block is constant. Which of the following graphs best describes how different forms of energy of the block vary with the distance travelled d during its upward journey?

A 

B 

C 

D 

Refer p.249, 252

13 Misc A 55 kg woman is standing on an escalator. The escalator moves with a constant velocity of 0.8 m s^{-1} and the moving direction makes an angle of 40° to the horizontal. Find the power delivered by the escalator on the woman.

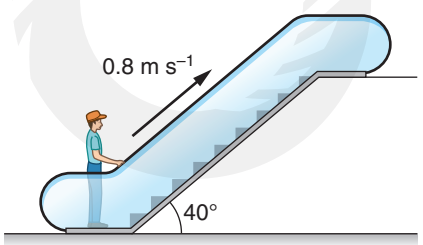


Fig h

A 277 W B 331 W

C 362 W D 432 W

Refer p.256

14 6.2 An object keeps moving forward along a straight line. Figure i shows how its KE varies with its displacement s when it travels along a straight line. The mass of the object is 2 kg. What is its acceleration?

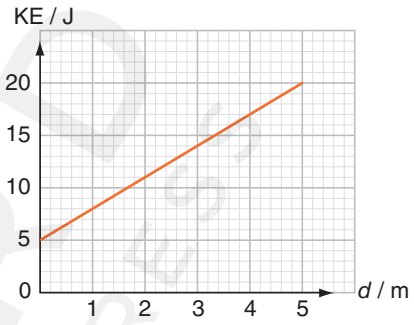


Fig i

A 1.5 m s^{-2} B 2 m s^{-2}

C 3 m s^{-2} D 6 m s^{-2}

Refer p.63, 237

15 6.3 HKDSE 2014 Paper 1A Q6

Two small identical blocks slide down from rest on smooth incline planes from the same height H as shown in Figure j and Figure k below. Their respective speeds at the bottom of the incline planes are denoted by v_1 and v_2 and the respective times taken to reach the bottom are t_1 and t_2. Which of the following is correct? Neglect air resistance.

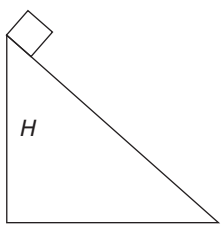


Fig j

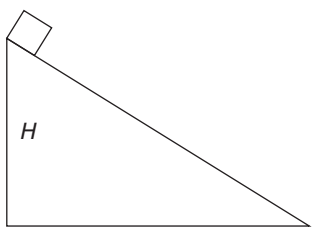


Fig k

A $v_1 > v_2$ and $t_1 = t_2$ (12%)

B $v_1 > v_2$ and $t_1 < t_2$ (33%)

C $v_1 = v_2$ and $t_1 = t_2$ (10%)

D $v_1 = v_2$ and $t_1 < t_2$ (45%)

Q15 Exam Report: Nearly half of the candidates wrongly chose the options that a block on a steeper incline plane of the same height would reach the bottom faster.

16 6.3 HKDSE 2015 Paper 1A Q8

An object falls from P to Q as shown. Throughout the motion, air resistance increases with the speed of the object.

Which of the following descriptions is/ are correct?

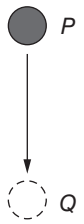


Fig l

(1) The net force acting on the object is constant throughout the motion.

(2) The magnitude of the object's acceleration decreases from P to Q.

(3) The kinetic energy gained by the object from P to Q is equal to its loss in gravitational potential energy.

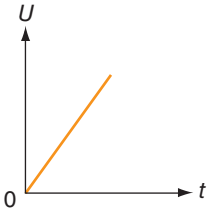
A (1) only B (2) only (47%)

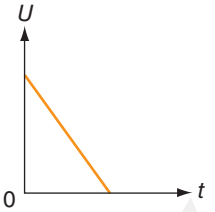
C (1) and (3) only D (2) and (3) only

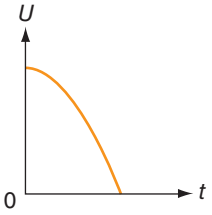
17 HKDSE 2016 Paper 1A Q8

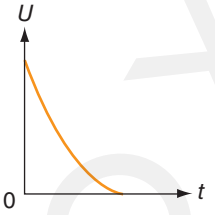
6.2

An object at a certain height falls freely from rest under gravity. Which graph correctly shows the variation of its gravitational potential energy U with time t? Neglect air resistance and take $U = 0$ at the ground.

A 

B 

C 

D 

18 HKDSE 2016 Paper 1A Q9

Misc

A crane moves a load of weight W steadily from point P to point Q as shown.

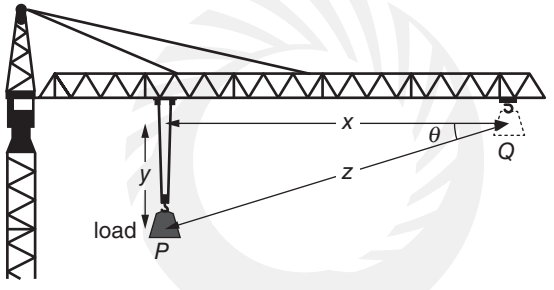


Fig m

The work done on the load by the crane is

A Wy. (25%)

B $W(x + y)$. (13%)

C Wz. (17%)

D $Wz \cos \theta$. (45%)

Q18 Exam Report: In this context, the work done by the crane turns into gain in gravitational potential energy. Only one quarter of the candidates were able to choose the correct answer.

Q19 Exam Report: About one-third of the candidates chose options which support the statement that the average output power of the machine was the same in both situations.

19 HKDSE 2017 Paper 1A Q11

Misc

A machine is fixed at the top of a smooth inclined plane. Two methods, (I) and (II), are used to lift a block from the ground to the top of the inclined plane by the machine.

(I) Pull the block vertically upward at a uniform speed v.

(II) Pull the block up along the inclined plane at the same uniform speed v.

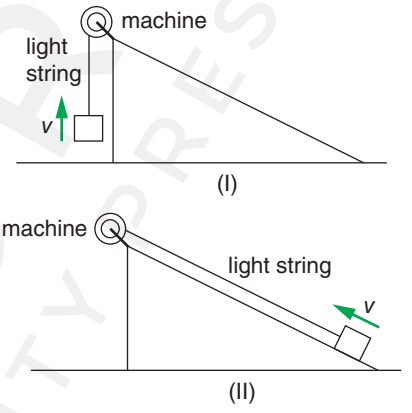


Fig n

Which of the following statements correctly compare(s) the two methods?

(1) The tension in the string is the same.

(2) The average output power of the machine is the same.

(3) The work done by the machine on the block is the same.

A (1) only (9%) B (3) only (57%)

C (1) and (2) only (9%) D (2) and (3) only (25%)

Conventional questions

★**20** In an air hockey game (Fig o), the puck is initially at rest and accelerates to 5 m s^{-1} after being hit by the mallet. The mass of the puck is 100 g. The friction acting on the puck is negligible.

6.3

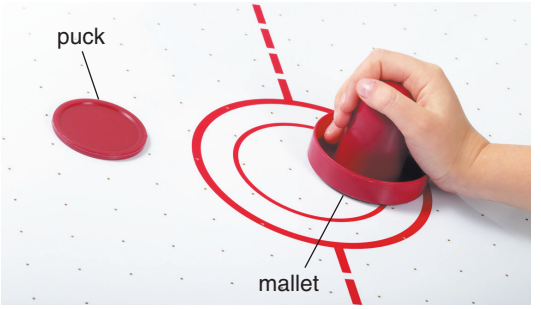


Fig o

(a) What is the work done by the mallet on the puck? (2 marks)

(b) Suppose the mallet pushes the puck over a distance of 1.2 cm. Find the average force acting on the puck by the mallet. (2 marks)

★21 6.3 Figure p shows a game in an amusement park. The boy hits the plank with a hammer and the metal cylinder moves upwards. He wins the game if the metal cylinder hits the bell 3 m above. The mass of the metal cylinder is 1 kg. The resistive force acting on the metal cylinder is 5 N.

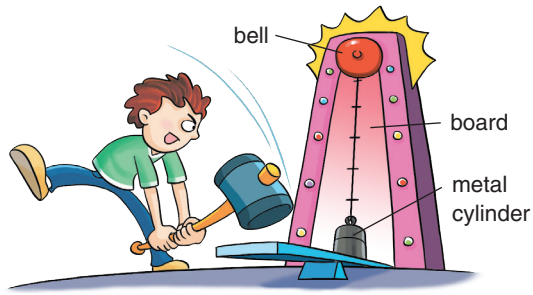


Fig p

(a) The boy strikes the plank and the metal cylinder leaves the plank at 4 m s^{-1}. Can the boy win the game? Show your calculation. (3 marks)

(b) Suggest two changes that would make it more difficult to win the game. (2 marks)

★22 Misc A motorcycle of mass 300 kg travels along a straight horizontal path (Fig q). It accelerates uniformly from 10 m s^{-1} to 25 m s^{-1} in 2 s. Assume that the resistive force acting on it is 900 N in this period.



Fig q

Q32 Exam Report: This question tested candidates' knowledge and understanding on work, energy and power using the example of a lift. Candidates' performance was satisfactory.

(b) (i) Only the more able candidates worked out correctly the total mechanical power output of the motor when the counterweight was installed.

(a) What is the distance travelled in this period? (2 marks)

(b) What is the average power output of the motorcycle's engine in this period? (2 marks)

(c) Is the power output of the motorcycle's engine constant in this period? Explain briefly. (3 marks)

★23 Misc A worker is loading goods onto a truck. He pushes a 60-kg box up an inclined plane at a uniform speed (Fig r). The plane is 3 m long and makes an angle of 20° to the horizontal. The friction between the box and the plane is 80 N.

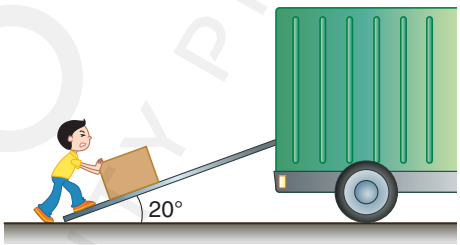


Fig r

(a) Find the work done on the box by the worker when the box reaches the end of the plane. (3 marks)

(b) Tom suggests using a longer plane which is inclined at a smaller angle to the horizontal (Fig s).

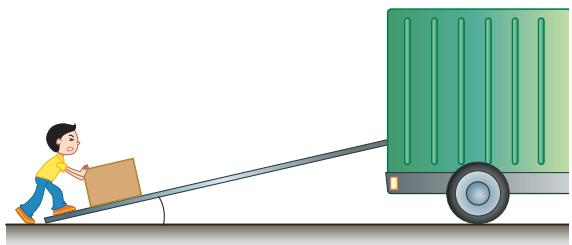


Fig s

(i) If the worker pushes the same box onto the truck using this new plane, is the work done by him the same as that in (a)? Explain your answer. (3 marks)

(ii) Suggest one advantage of Tom's suggestion. (1 mark)

(ii) Most knew that the output power required from the motor was smaller with the addition of the counterweight.

(iii) Few candidates were able to explain clearly why the claim was not true. Some wrongly thought that the lift could still be raised even when the drum's surface was frictionless.

★24 Misc Two blocks X and Y, which are initially at rest on a smooth horizontal surface, are pulled by the same horizontal force F provided by motors. X has a larger mass than Y. The motor pulling each block stops after each block has moved by the same displacement s.

(a) Compare the final KE of the blocks. Explain your answer. (2 marks)

(b) Compare the final speed of the blocks. Explain your answer. (2 marks)

(c) Compare the average power delivered by the motor on the blocks. Explain your answer. (3 marks)

★25 6.3 Felix Baumgartner went skydiving from a height 39 km above the ground (Fig t). He fell from rest and reached a terminal speed of 377 m s^{-1}. When he was 1500 m above the ground, he opened his parachute and then landed safely. Assume that g = 9.81 m s^{-2} in the whole process.



Fig t

(a) (i) If the resistance was negligible, estimate the distance he travelled before his speed reached 377 m s^{-1}. (2 marks)

(ii) In fact, air resistance is not negligible. How would the answer to (i) be affected when air resistance is taken into account? Explain your answer in terms of energy conversion. (2 marks)

(b) Suppose his total mass was 100 kg and his speed was 377 m s^{-1} when he opened the parachute. Estimate the total work done against air resistance from the start of the jump to the moment he opened the parachute. (2 marks)

(c) Suppose he landed at a speed of 2 m s^{-1}. Estimate the average air resistance acting on him after the parachute was opened. (2 marks)

★26 6.3 Fanny goes bungee jumping from a bridge 20 m above a river. Her body is connected to the bridge with an elastic string. When Fanny falls from rest to point A which is 8 m below the bridge, the elastic string starts to extend. The mass of Fanny is 50 kg.

(a) Find the loss in Fanny's PE as she falls from the bridge to A. (2 marks)

(b) When Fanny reaches the lowest point B, she is momentarily at rest and then bounces up. B is 12 m below A. Suppose the total mechanical energy of Fanny and the string remains constant.

(i) What is the average force acting on her by the string during her fall from A to B? (2 marks)

(ii) 'The net force acting on Fanny is zero at B.' Comment on this statement. (2 marks)

(c) In reality, Fanny stops moving after bouncing up and down several times. Suggest where the energy goes. (1 mark)

★27 Misc A meteor is formed when a rock from space enters the Earth's atmosphere at a very high speed (Fig u).



Fig u

Suppose a rock of mass 2 kg enters the Earth's atmosphere at 20 km s^{-1}. It slows down to 19 km s^{-1} after a while. Assume that its mass remains unchanged and its change in PE is negligible compared with its change in KE within this period.

(a) Find the KE lost by the rock. (2 marks)

(b) If one-fifth of the energy in (a) becomes the internal energy of the rock, estimate the rise in temperature of the rock. Take the specific heat capacity of the rock as 2000 J kg^{-1} $°C^{-1}$. (2 marks)

★28 6.3 Peter is investigating the friction of a rough surface using the following experiment. He releases a box of 5 kg at A and allows it to move down from rest along a smooth slope AB (Fig v). The box then moves along a rough horizontal surface BC and eventually stops. Peter measures the stopping distance s on the rough surface. A is 2 m above the rough surface.

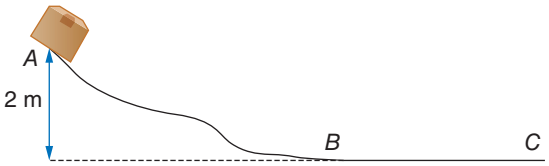


Fig v

(a) Find the speed of the box when it reaches B. (2 marks)

(b) If s is 10 m, estimate the friction acting on the box by the rough surface. (2 marks)

(c) Mary repeated the experiment using another box with a larger mass. If the friction acting on the box by the rough surface is assumed to be unchanged, will s be expected to become larger than, smaller than or equal to the value in (b)? Explain your answer. (3 marks)

★★29 Misc A helicopter is stationary above the sea. A man of mass 70 kg is pulled from the sea up to the helicopter by a rope. Figure w shows the variation of the speed v of the man with time t.

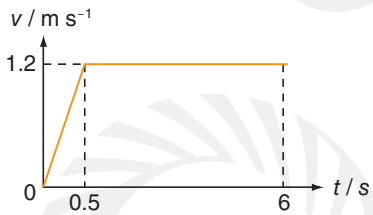


Fig w

(a) Find the magnitude of the tension in the rope at t = 0.3 s. (3 marks)

(b) Find the work done by the tension on the man from t = 0.5 s to t = 6 s. (3 marks)

(c) Sketch a graph to show how the power P delivered by the tension varies with t from t = 0 to t = 6 s. What is the meaning of the area under this graph? (4 marks)

Refer p.262

★★30 Misc Two identical balls P and Q slide down along two smooth tracks of the same length from rest from the same height as shown (Fig x).

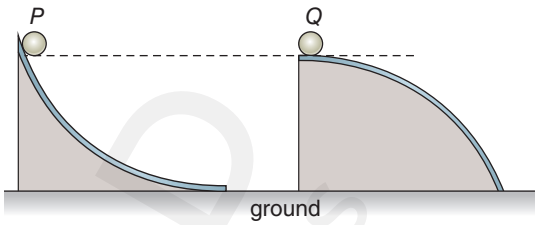


Fig x

(a) Which ball gains more KE when reaching the ground? Explain briefly. (2 marks)

(b) (i) Sketch a graph to show how the speed of each ball varies with time. (2 marks)

(ii) Which ball reaches the ground first? (1 mark)

Refer p.247

★★31 6.3 Using a football, a wireless motion sensor and a device with a data logging program, describe an experiment to investigate the law of conservation of energy. You should include the physical quantities to be measured and the result expected. (5 marks)

Refer p.246

See p.271 for the Exam Report.

32 HKDSE 2013 Paper 1B Q3 6.4

A lift car of weight 8000 N is going up with constant speed 2 m s^{-1} as shown in Figure y. The upward force raising the lift car is provided by the cable wound on a drum which is driven by a motor. The other end of the cable is firmly attached to the drum at P. Neglect air resistance and the mass of the cable.

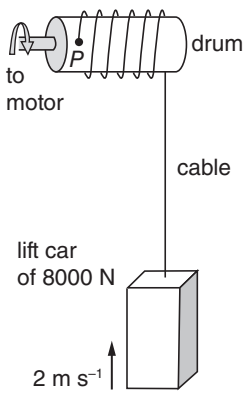


Fig y

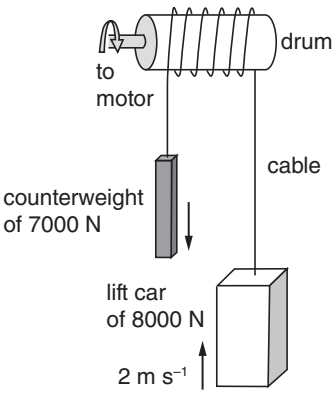


Fig z

(a) (i) Calculate the mechanical power delivered to the rising lift car by the motor. (2 marks)

(ii) The total mechanical power output of the motor is 20 kW. How much power is lost due to overcoming friction between the movable parts? (1 mark)

(b) Now a 7000 N counterweight is installed at the other end of the cable as shown in Figure z. The counterweight always moves in the opposite direction to the lift car which again moves up at 2 m s^{-1}. Assume that there is no slipping between the cable and the drum.

(i) Calculate the total mechanical power output of the motor required in this case, assuming the same power loss in overcoming friction between movable parts as found in (a). (2 marks)

(ii) State the advantage of having the counterweight installed. (1 mark)

(iii) A claim is made that as power is lost due to friction, a drum with frictionless surface can further reduce the power required from the motor. Comment on this claim. (2 marks)

33 HKDSE 2013 Paper 1B Q5

6.3

Figure aa shows a smooth sloping track *ABC* firmly fixed in a vertical plane with its horizontal part *BC* resting on a bench surface. You are given a toy skier, a metre rule and a long rough paper strip with adhesive tape on the bottom surface.

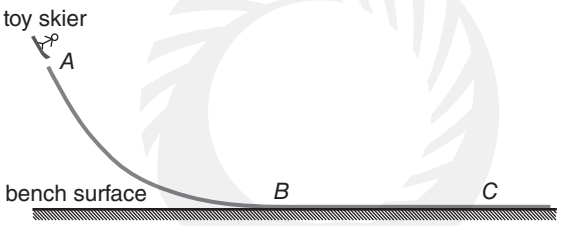


Fig aa

Using the apparatus provided, describe an experiment to study how the stopping distance of the toy skier depends on its height of release. Your description should include the physical quantities to be measured and the result expected. (5 marks)

34 HKDSE 2020 Paper 1B Q4

6.3

A small sphere is released from rest at point *A* and runs along a smooth track *ABC* as shown in Figure ab. The track around the lowest point *B* is approximately circular in shape.

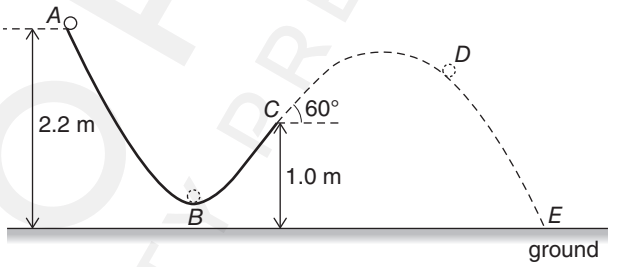


Fig ab

The sphere leaves the track at point *C* where the track makes an angle of 60° with the horizontal. It finally reaches point *E* on the ground. Neglect air resistance. (g = 9.81 m s^{-2})

(a) Arrange the speeds of the sphere at points *A*, *B*, *C* and *D* respectively in descending order. (1 mark)

(b) On Figure ab, use arrows to indicate the acceleration of the sphere, if any, at point *B* and at point *D* respectively. (2 marks)

(c) (i) Describe the energy conversion of the sphere when it goes along the track *ABC*. (2 marks)

(ii) Hence find the speed of the sphere at point *C*. (2 marks)

Q33 Exam Report: This question required candidates to use the apparatus provided to describe an experiment to study how the stopping distance of a toy skier depends on its height of release. The general performance was poor. Only the more able candidates showed clearly in their answers the procedures taken, the physical quantities to be measured and the expected result based on the work-energy principle. Many candidates stated that the stopping distance would increase with the height of release, but without any graphical or mathematical elaboration. Quite a number of them did not make proper use of the long rough paper strip and the metre rule provided to perform the experiment.

Q34 Exam Report: Candidates' performance was satisfactory.

(a) Some candidates wrongly arranged the speeds in the opposite order.

(b) A lot of the candidates confused the following: acceleration, weight, acceleration due to gravity and velocity.

(c) (i) Most were able to describe the energy conversion correctly. However, some failed to point out that the energy conversion from *B* to *C* only involved '*part* of the sphere's kinetic energy'.

(ii) This part was well answered. Candidates were able to use energy conservation to solve the problem.

35 HKDSE 2018 Paper 1B Q3
6.3

Read the following passage about **braking in vehicles** and answer the questions that follow.

Mechanical braking is the most common braking method in vehicles. In this method, wheels of vehicles are fitted with brake pads which apply frictional forces that inhibit the motion of the wheels. Frictional braking results in a conversion of the vehicle's kinetic energy to thermal energy which then dissipates to the atmosphere in the form of waste heat.

The thermal energy (in J) generated during braking is approximately given by:

$$\text{thermal energy generated} = F_r \times d$$

where F_r is the frictional forces (in N), and

d is the braking distance (in m).

Applying the principle of energy conservation:

kinetic energy dissipated = thermal energy generated

$$\frac{1}{2}mv^2 = F_r \times d$$

From this equation it can be seen that increasing the velocity v or mass m of an object means the applied frictional forces must be increased in order to bring the object to a stop within the same distance.

(a) Referring to the passage, explain why there should be a maximum load limit for a vehicle (see the vehicle label example below). (2 marks)

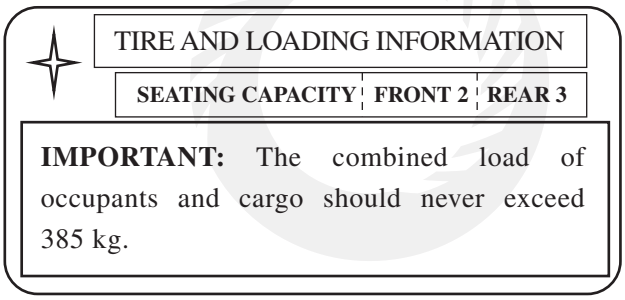


Fig ac

(b) (i) It is known that the brake pads would provide much smaller frictional forces or even brake failure would occur (i.e. without any friction provided) when reaching a temperature of over a few hundred degrees Celsius. Explain why it is not recommended for a vehicle driver to **apply the brakes continuously** during a long downhill trip. (1 mark)

(ii) Figure ad shows an emergency escape ramp (slanting upwards) built for stopping vehicles with brake failure resulting from the situation described in (b)(i). If such a ramp makes an angel of 30° with the horizontal and a vehicle with brake failure enters the bottom end of the ramp at a speed of 25 m s^{-1}, estimate how far it will travel along the ramp before it stops. Neglect air resistance and mechanical resistances within the vehicle. (g = 9.81 m s^{-2}) (2 marks)



Fig ad

Q35 Exam Report: Candidates' performance was satisfactory.

(a) Many candidates missed the point that if the friction F_r remained unchanged (at its maximum value) while the maximum load limit was exceeded, the braking distance d would be longer according to the equation $\frac{1}{2}mv^2 = F_r \times d$ given.

(b) (i) When explaining why it was not recommended to apply the brakes continuously, only the more able ones pointed out that the thermal energy generated would cause the temperature to reach over a few hundred degrees Celsius as mentioned in the passage.

(ii) Some candidates had difficulties estimating how far the vehicle would travel up the ramp by applying either the principle of energy conservation or the equations for uniformly decelerated motion. A few even did not know that the deceleration up the ramp was $g \sin \theta$.

Know more

Holding a heavy object still (p.232)

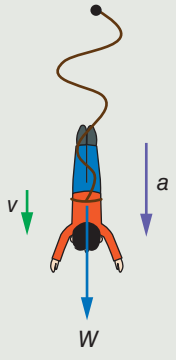
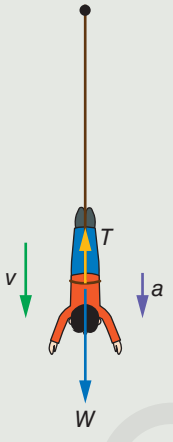
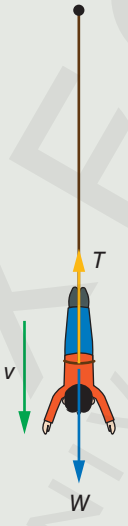
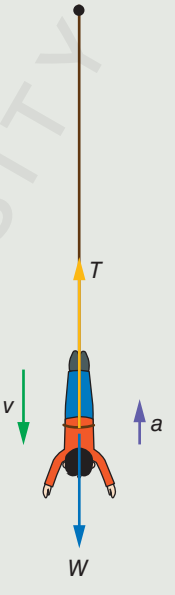
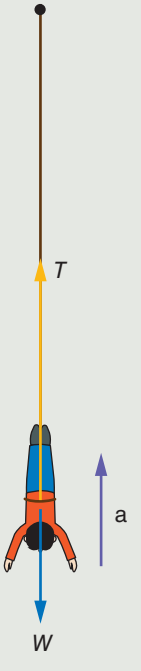
We feel tired after pushing a heavy box for a distance. We also feel tired after holding a heavy box still for some time. However, in the latter case, we **do not do work on the box!** We are tired because our muscle fibres contract and relax repeatedly throughout the process. This microscopic work consumes energy and makes us feel tired.



Know more

Motion in bungee jumping (p.250)

The downward motion of bungee jumping can be divided into several stages. Note that the tension T in the string increases with the extension.

				
The string is not yet stretched. F_{net} is equal to W as T is zero.	The string starts stretching. T increases with the extension. Since $F_{net} = W - T$, F_{net} decreases.	When the person falls to a certain point, $T = W$ and $F_{net} = W - T = 0$.	When the person continues to fall, T keeps increasing and is larger than W. F_{net} points upwards and so the person slows down.	The person slows down until he is momentarily at rest at the lowest position. Both T and F_{net} are maximum at this instant.
$T = 0$	$T < W$	$T = W$	$T > W$	T max
$F_{net} = W$ (downwards)	$F_{net}\downarrow$ (downwards)	$F_{net} = 0$	$F_{net}\uparrow$ (upwards)	F_{net} max (upwards)
$a = g$ (downwards)	$a\downarrow$ (downwards)	$a = 0$	$a\uparrow$ (upwards)	a max (upwards)
$v\uparrow$ (downwards)	$v\uparrow$ (downwards)	v max (downwards)	$v\downarrow$ (downwards)	$v = 0$

Note that $\uparrow$ means increasing and $\downarrow$ means decreasing